EN.553.733 Nonparametric Bayesian Statistics

Jonathan Y. Ma

June 2026

This course was taught by Dr. Yanxun Xu in the Spring 2026 Semester. The notes are transcribed by Jonathan Ma and may contain errors and omitted content, so any issues are to be directed towards me.

Contents

1 Introduction **7**
- 1.1 Exchangeability and De Finetti's Theorem . . . 7
- 1.2 The Bayesian Toolkit . . . 8
  - 1.2.1 Likelihood . . . 8
  - 1.2.2 Prior Distribution . . . 8
  - 1.2.3 Joint, Marginal, and Posterior . . . 8
  - 1.2.4 Predictive Distribution . . . 9
- 1.3 Conjugate Priors . . . 9
  - 1.3.1 Definition . . . 9
  - 1.3.2 Properties . . . 9
- 1.4 Credible Intervals . . . 9
- 1.5 MAP Estimator . . . 10
- 1.6 Testing Hypotheses . . . 10
- 1.7 Bayes Factor . . . 10
- 1.8 Jeffreys' Scale . . . 11
- 1.9 Markov Chain Monte Carlo (MCMC) . . . 11
- 1.10 Additional Remarks . . . 12
- 1.11 Gibbs Sampling . . . 13
  - 1.11.1 Overview . . . 13
  - 1.11.2 Key Idea . . . 13
  - 1.11.3 Two-Stage Gibbs Sampler . . . 13
  - 1.11.4 Multi-Stage Gibbs Sampler . . . 13
  - 1.11.5 Remarks on Gibbs Sampling . . . 14
  - 1.11.6 Convergence Diagnostics . . . 14
- 1.12 Autocorrelation . . . 14
- 1.13 Reading: Mixture Models and Bayesian Computation . . . 15

2 Dirichlet Process **18**
- 2.1 Nonparametric Bayesian Priors . . . 18
- 2.2 Finite Mixture Model . . . 19
- 2.3 Beta Distribution Review . . . 19
- 2.4 Dirichlet Distribution . . . 20
- 2.5 Finite Mixture Model (General K) . . . 21

2.6 When $K > N$ . . . 21
2.7 Choosing $K = \infty$ . . . 21
2.8 Stick-Breaking Construction . . . 22
2.8.1 Finite Construction . . . 22
2.8.2 Infinite Stick-Breaking . . . 22
2.8.3 GEM Distribution . . . 23
2.9 Interpretation . . . 23
2.10 Readings: Foundations of the Dirichlet Process . . . 23
2.10.1 Dirichlet Process Mixture Model (DPMM) . . . 26
2.10.2 Pólya Urn Interpretation . . . 30
2.10.3 Blackwell–MacQueen Pólya Urn . . . 31
2.11 Posterior Support, Consistency, and Contraction . . . 32
2.12 Chinese Restaurant Process (CRP) . . . 33
2.13 Number of Clusters . . . 35
2.13.1 Expected Value . . . 35
2.13.2 Variance . . . 35
2.13.3 Distribution of K_n . . . 36
2.13.4 Interpretation . . . 36

3 Sampling Techniques **37**
3.1 Sampling Techniques for DP Mixture Models . . . 37
3.1.1 Finite Mixture Models . . . 38
3.1.2 Bayesian Mixture Model . . . 38
3.1.3 Gibbs Sampling (Finite Case) . . . 38
3.2 CRP Mixture Model . . . 38
3.3 DP Mixture Model (DPM) . . . 39
3.4 Alternative Representations . . . 39
3.4.1 Stick-Breaking Form . . . 39
3.4.2 Clustering Representation . . . 39
3.5 Collapsed Gibbs Sampling . . . 40
3.5.1 Collapsed assignment probabilities . . . 40
3.5.2 Cluster Parameter Update . . . 40
3.6 Gaussian DPM Example . . . 40
3.7 Reading: Mixture of Dirichlet Process Models . . . 41
3.8 Reading: Sampling Methods for Dirichlet Process Mixtures . . . 42
3.9 Sampling Techniques: Slice Sampling for DP Mixtures . . . 44
3.9.1 Reading: Gibbs Sampling for Stick-Breaking Priors . . . 49
3.10 Reading: Slice Sampling Mixture Models . . . 51
3.11 Reading: Bayesian Density Estimation via DP Mixtures . . . 52
3.12 Advanced MCMC Methods . . . 53
3.12.1 Random Walk Metropolis–Hastings (RWMH) . . . 53
3.12.2 Metropolis Adjusted Langevin Algorithm (MALA) . . . 54
3.13 Auxiliary Variable Methods and Hybrid MCMC . . . 56
3.13.1 Adaptive Gibbs Samplers . . . 57
3.13.2 Random Scan Gibbs Sampler . . . 57

3.13.3 Adaptive Metropolis-within-Gibbs (Full) . . . 59
3.14 Pitman–Yor Process (PYP) . . . 60

4 Extensions of the Dirichlet Process 62
4.1 Pitman–Yor Process: Inference and Applications . . . 62
4.1.1 Inference under Pitman–Yor Process . . . 62
4.1.2 Case Study: Genetics . . . 63
4.1.3 Case Study: Topic Modeling . . . 63
4.2 Hierarchical Dirichlet Process (HDP) . . . 64
4.3 Hierarchical Dirichlet Process: Constructions and Inference . . . 65
4.3.1 Stick-Breaking Construction of HDP . . . 65
4.3.2 Chinese Restaurant Franchise (CRF) . . . 66
4.3.3 Case Study: BNP Modeling of Marathon Data . . . 68
4.4 Dependent Dirichlet Processes (DDP) . . . 68
4.4.1 Dependent Dirichlet Processes (DDP) . . . 70
4.4.2 Weighted Mixture of Dirichlet Processes . . . 71
4.4.3 Kernel Stick-Breaking Process (KSBP) . . . 71
4.4.4 Predictor-Dependent Clustering . . . 71
4.4.5 Example: Information Retrieval . . . 72
4.4.6 Nested Dirichlet Process . . . 72
4.4.7 ANOVA Dependent Dirichlet Process (ANOVA-DDP) . . . 72
4.4.8 Problem Setting for BNP Regression . . . 73
4.4.9 Gaussian Process (GP) . . . 73
4.4.10 Common Covariance Functions . . . 73
4.4.11 Combining Kernels . . . 74
4.4.12 GP Regression . . . 74
4.4.13 Prediction (Multi-step) . . . 75
4.4.14 Learning Hyperparameters . . . 75
4.4.15 Remarks . . . 75
4.4.16 Feature Selection . . . 75
4.4.17 Feature Selection in Bayesian Models . . . 76
4.4.18 Limitations of Gaussian Processes . . . 76
4.4.19 Scalable Gaussian Processes . . . 76
4.4.20 CART: Regression Trees . . . 77
4.4.21 Bayesian Regression Trees . . . 77
4.4.22 Bayesian Additive Regression Trees (BART) . . . 77
4.4.23 Regression Tree Representation . . . 77
4.4.24 MCMC for Tree Models . . . 78
4.4.25 Pros and Cons . . . 78
4.5 Random Partition Models . . . 78
4.5.1 Inference on Partitions . . . 78
4.5.2 Sampling Model . . . 79
4.5.3 Product Partition Model (PPM) . . . 79
4.5.4 Species Sampling Models (SSM) . . . 79
4.5.5 Covariate-Dependent Clustering . . . 80

4.5.6 DDP-like Constructions . . . 81
4.5.7 Item Allocation and Attraction-Based Clustering . . . 82
4.5.8 PPM with Covariates (PPMx) . . . 82
4.5.9 Example: Mixed Effects Model . . . 83

5 Latent Feature Models and the IBP 85
5.1 Infinite Limit Construction . . . 85
5.2 Indian Buffet Process (IBP) . . . 86
5.2.1 Posterior Inference . . . 86
5.2.2 Properties of IBP . . . 86
5.3 Two-Parameter Extension . . . 87
5.4 IBP from the Beta–Bernoulli Process . . . 88
5.5 Hierarchical Beta Process . . . 88
5.6 Completely Random Measures (CRM) . . . 88
5.6.1 Lévy–Khintchine Representation . . . 89
5.7 Poisson Point Process . . . 89
5.8 Gamma Process . . . 89
5.8.1 Connection to Dirichlet Process . . . 89
5.9 Beta Process . . . 90
5.10 Interpretation Across Processes . . . 90
5.11 Example: Linear Gaussian Latent Feature Model . . . 90
5.12 From IBP to Count-Valued Feature Models . . . 92
5.13 Negative Binomial Factor Analysis (NBFA) . . . 92
5.14 Double Feature Allocation (DFA) . . . 92
5.15 Latent Disease Interpretation . . . 92
5.16 Patient–Disease (PD) Model . . . 92
5.17 Symptom–Disease (SD) Model . . . 93
5.18 Sampling Model . . . 93
5.19 Scalability Challenges . . . 93
5.20 Consensus Monte Carlo . . . 93
5.21 Modified Consensus Monte Carlo . . . 94

6 Polya Tree Priors 95
6.1 Prior Construction . . . 95
6.2 Ingredients of a Polya Tree . . . 95
6.3 Recursive Partition Tree . . . 95
6.3.1 Binary Indexing . . . 95
6.4 Polya Tree Definition . . . 96
6.5 Tailfree Processes . . . 96
6.6 Dirichlet Process as a Special Case . . . 96
6.6.1 Consistency Property . . . 97
6.7 Interpretation . . . 97
6.8 Absolute Continuity of Polya Trees . . . 97
6.9 Effect of Parameters . . . 98
6.10 Prior Centering . . . 98

6.11 Centering Method 1: Quantile Partition . . . 98
6.12 Centering Method 2: Match Split Probabilities . . . 98
6.13 Why This Centers the Prior . . . 98
6.14 Posterior Inference . . . 99
6.15 Posterior Predictive Sampling . . . 99
6.16 Marginal Distribution . . . 99
6.17 Polya Tree: Marginal Distribution . . . 100
6.18 Mixture of Polya Trees (MPT) . . . 100
6.19 Motivation: Joint Modeling . . . 100
6.20 Multivariate Polya Tree . . . 100
6.21 Rubbery Polya Tree . . . 101
6.22 Linear Dependent Tailfree Process (LDTP) . . . 101
6.23 Why LDTP Helps . . . 101
6.24 LDTP Case Study: Growth Curve Analysis . . . 101
6.25 Dependent Multivariate Polya Trees . . . 102
6.26 Multivariate Beta Process (MBP) . . . 102
6.27 MBP Construction . . . 102
6.28 Finite-Dimensional Representation for MCMC . . . 103
6.29 Mixture of Polya Trees Revisited . . . 103
6.30 Polya Trees in Joint Modeling . . . 103
6.31 Summary of Extensions . . . 103
6.32 Alternative Construction via MCMC . . . 103
6.33 DP Representation . . . 104
6.34 Polya Urn Sampling . . . 104
6.35 Take-Home Message . . . 104

7 BNP Survival Analysis 105
7.1 Survival Analysis: Setup . . . 105
7.2 Semiparametric Survival Models . . . 105
7.2.1 Proportional Hazards (PH) . . . 105
7.2.2 Accelerated Failure Time (AFT) . . . 106
7.2.3 Proportional Odds (PO) . . . 106
7.3 Bayesian Semiparametric PH Models . . . 106
7.4 Neutral-to-the-Right (NTR) Processes . . . 106
7.5 Fully Nonparametric Survival Regression . . . 107
7.5.1 ANCOVA-DDP . . . 107
7.5.2 Linear Dependent Tailfree Process (LDTP) . . . 107
7.6 Why BNP Survival Models . . . 107
7.7 Fully Nonparametric Survival Regression: LDTP (Interpretation) . . . 107
7.7.1 Interpretation I . . . 107
7.7.2 Interpretation II . . . 108
7.8 Alternative BNP Construction: DDP-GP . . . 108
7.8.1 Model Construction . . . 108
7.8.2 Interpretation . . . 108
7.9 DDP-GP Model for Survival Data . . . 108

7.10 Comparison of BNP Survival Models 109

Chapter 1

Introduction

For observed data y and an unknown parameter θ, Bayes' rule gives

$$p(\theta \mid y) \propto p(y \mid \theta)p(\theta).$$

Bayesian nonparametrics extends this framework by allowing the effective model dimension to grow with the data. The parameter may be a probability measure, a function, a partition, or an infinite feature allocation rather than a fixed finite-dimensional vector.

1.1 Exchangeability and De Finetti's Theorem

A theoretical motivation for nonparametric Bayesian methods comes from *De Finetti's Theorem*.

Exchangeability

A sequence $(X_n)_{n\geq 1}$ is *infinitely exchangeable* if for every N and every permutation σ of $\{1, \ldots, N\}$,

$$p(X_1, \ldots, X_N) = p\Big(X_{\sigma(1)}, \ldots, X_{\sigma(N)}\Big).$$

This means the joint distribution depends only on the multiset of observations, not their order.

De Finetti Representation

Theorem 1.1.1 (De Finetti (informal))**.** *A sequence $X_1, X_2, \ldots$ is infinitely exchangeable if and only if, for all N, there exists a probability measure P such that*

$$p(X_1, \ldots, X_N) = \int_\Theta \prod_{n=1}^{N} p(X_n \mid \theta)\, P(d\theta).$$

Conditionally on θ, the observations are i.i.d., i.e.,

$$X_n \mid \theta \overset{i.i.d.}{\sim} p(\cdot \mid \theta).$$

Thus exchangeability implies a mixture-of-i.i.d. representation.

Implications

The representation separates the sampling law $p(x \mid \theta)$ from the mixing distribution $P(d\theta)$. In a parametric model, P is a prior on a finite-dimensional parameter. In a nonparametric Bayesian model, the random object mixed over may itself be infinite-dimensional, while exchangeability still supplies the conditional-i.i.d. interpretation.

1.2 The Bayesian Toolkit

1.2.1 Likelihood

At the core of Bayesian inference is the *likelihood*. Given data $D = x = (x_1, \ldots, x_n)$,

$$L(\theta \mid D) = f(x \mid \theta).$$

If $x_1, \ldots, x_n$ are conditionally independent given θ, then

$$f(x \mid \theta) = \prod_{i=1}^{n} f(x_i \mid \theta).$$

1.2.2 Prior Distribution

Uncertainty about θ is encoded through a *prior* $\pi(\theta)$.

1.2.3 Joint, Marginal, and Posterior

Joint distribution

$$\psi(\theta, x) = f(x \mid \theta)\pi(\theta).$$

Marginal distribution

$$m(x) = \int f(x \mid \theta)\pi(\theta)\, d\theta.$$

This quantity is also called the *evidence* or *model likelihood*.

Posterior distribution

$$\pi(\theta \mid x) = \frac{f(x \mid \theta)\pi(\theta)}{\int f(x \mid \theta)\pi(\theta)\, d\theta} = \frac{f(x \mid \theta)\pi(\theta)}{m(x)}.$$

1.2.4 Predictive Distribution

If $y \sim g(y \mid \theta, x)$, then the predictive distribution is

$$g(y \mid x) = \int g(y \mid \theta, x)\, \pi(\theta \mid x)\, d\theta.$$

This integrates parameter uncertainty into predictions.

Remark While inference focuses on $\pi(\theta \mid x)$, prediction via $g(y \mid x)$ is often the primary goal.

1.3 Conjugate Priors

1.3.1 Definition

Theorem 1.3.1 (Conjugate Prior)**.** *A class $\mathcal{P}$ of prior distributions is conjugate for a likelihood $p(x \mid \theta)$ if*

$$\pi(\theta) \in \mathcal{P} \;\Rightarrow\; \pi(\theta \mid x) \in \mathcal{P}.$$

Examples

If $x \mid \theta \sim \text{Bin}(n, \theta)$ and $\theta \sim \text{Beta}(a, b)$, then

$$\theta \mid x \sim \text{Beta}(a + x,\ b + n - x).$$

If $x_1, \ldots, x_n \overset{i.i.d.}{\sim} \text{Poisson}(\lambda)$ and $\lambda \sim \text{Gamma}(a, b)$, then

$$\lambda \mid x \sim \text{Gamma}\Big(a + \textstyle\sum_{i=1}^{n} x_i,\ b + n\Big).$$

1.3.2 Properties

Advantage *Closed-form* posterior updates.

Disadvantage The conjugate family may not reflect prior beliefs.

1.4 Credible Intervals

Theorem 1.4.1 (Credible Interval)**.** *An interval (a, b) is a $(1-\alpha)100\%$ credible interval for θ if*

$$\Pr(a < \theta < b \mid x) = 1 - \alpha.$$

Quantile Construction

Choose a, b such that

$$\Pr(\theta < a \mid x) = \frac{\alpha}{2}, \qquad \Pr(\theta > b \mid x) = \frac{\alpha}{2}.$$

Remark

Credible intervals quantify posterior uncertainty, unlike frequentist confidence intervals which are defined via repeated sampling.

1.5 MAP Estimator

The *maximum a posteriori (MAP)* estimator is defined as

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} f(x \mid \theta)\pi(\theta).$$

Equivalently,

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} \left[\log f(x \mid \theta) + \log \pi(\theta)\right].$$

Motivation

The MAP estimator is a penalized-likelihood estimator because $\log \pi(\theta)$ acts as a penalty. It can be useful when optimization is much easier than integration, but it does not represent posterior uncertainty and is not invariant to nonlinear reparameterization.

1.6 Testing Hypotheses

Hypothesis testing on θ is formulated as

$$H_0 : \theta \in \Theta_0 \qquad \text{versus} \qquad H_1 : \theta \notin \Theta_0.$$

1.7 Bayes Factor

Bayesian testing procedures can be based on the posterior probability

$$P^{\pi}(\theta \in \Theta_0 \mid x),$$

or alternatively on the *Bayes factor*.

Definition

The Bayes factor comparing Θ_1 to Θ_0 is

$$B_{10}^{\pi} = \frac{P^{\pi}(\theta \in \Theta_1 \mid x)/P^{\pi}(\theta \in \Theta_0 \mid x)}{P^{\pi}(\theta \in \Theta_1)/P^{\pi}(\theta \in \Theta_0)}.$$

Model Comparison Formulation

For corresponding models $\mathcal{M}_1$ versus $\mathcal{M}_0$,

$$B_{10}^{\pi} = \frac{P^{\pi}(\mathcal{M}_1 \mid x)}{P^{\pi}(\mathcal{M}_0 \mid x)} \Big/ \frac{P^{\pi}(\mathcal{M}_1)}{P^{\pi}(\mathcal{M}_0)}.$$

If the prior is written as

$$\pi(\theta) = \Pr(\theta \in \Theta_1)\, \pi_1(\theta) + \Pr(\theta \in \Theta_0)\, \pi_0(\theta),$$

then

$$B_{10}^{\pi} = \frac{\int f(x \mid \theta_1)\pi_1(\theta_1)\, d\theta_1}{\int f(x \mid \theta_0)\pi_0(\theta_0)\, d\theta_0} = \frac{m_1(x)}{m_0(x)}.$$

Thus the Bayes factor is *akin to a likelihood ratio*.

1.8 Jeffreys' Scale

Interpretation of the Bayes factor is often based on $\log_{10}(B_{10}^{\pi})$:

if $\log_{10}(B_{10}^{\pi}) \in [0, 0.5]$, evidence against H_0 is poor; if $\log_{10}(B_{10}^{\pi}) \in [0.5, 1]$, it is substantial; if $\log_{10}(B_{10}^{\pi}) \in [1, 2]$, it is strong; if $\log_{10}(B_{10}^{\pi}) > 2$, it is decisive.

1.9 Markov Chain Monte Carlo (MCMC)

Statistical inference is based on the posterior distribution

$$\pi(\theta \mid x) = \frac{\pi(\theta) f(x \mid \theta)}{\int \pi(\theta) f(x \mid \theta)\, d\theta}.$$

Motivation

Conjugate models permit analytic posterior integration, but most realistic hierarchical models do not. Simulation is then used to approximate posterior expectations, credible intervals, predictive distributions, and testing functionals from a common collection of dependent draws.

The MCMC Principle

Definition

An *MCMC method* for simulating from a distribution π is any method that constructs an ergodic Markov chain (X_n) whose stationary distribution is π.

Given an initial value $x^{(0)}$, the chain is generated via a transition kernel such that

$$X_n \xrightarrow{d} X \sim \pi.$$

Construction

Metropolis–Hastings constructs a reversible accept–reject kernel, whereas Gibbs sampling composes draws from full conditional distributions. Diagnostics cannot certify convergence at a finite time; they assess whether multiple runs show evidence of nonstationarity, poor mixing, or failure to explore relevant regions.

Metropolis–Hastings Algorithm

Algorithm

Given current state $x^{(t)}$:

Generate proposal $Y_t \sim q(y \mid x^{(t)})$ (*proposal distribution*). Set

$$X^{(t+1)} = \begin{cases} Y_t, & \text{with probability } \rho(x^{(t)}, Y_t), \\ x^{(t)}, & \text{with probability } 1 - \rho(x^{(t)}, Y_t), \end{cases}$$

where

$$\rho(x, y) = \min\left\{ \frac{\pi(y)\, q(x \mid y)}{\pi(x)\, q(y \mid x)},\ 1 \right\}.$$

Remarks on Metropolis–Hastings

The quantity $\rho(x, y)$ is the *acceptance ratio*. The resulting chain is Markov since $X^{(t+1)}$ depends only on $X^{(t)}$. If $q(x \mid y) = q(y \mid x)$, the proposal is symmetric and the method reduces to the *Metropolis algorithm*. In this case, the acceptance probability simplifies to

$$\frac{\pi(y_t)}{\pi(x^{(t)})}.$$

1.10 Additional Remarks

Acceptance Mechanism

If

$$\frac{\pi(y_t)}{q(y_t \mid x^{(t)})} > \frac{\pi(x_t)}{q(x^{(t)} \mid y_t)},$$

the proposal is always accepted. Otherwise, it is accepted with probability ρ.

Invariance

The algorithm depends only on ratios:

$$\frac{\pi(y_t)}{\pi(x^{(t)})}, \qquad \frac{q(x^{(t)} \mid y_t)}{q(y_t \mid x^{(t)})}.$$

Thus, normalizing constants cancel out, provided they do not depend on the conditioning variables.

Choice of Proposal

The choice of proposal distribution q strongly affects performance (efficiency and convergence).

A good proposal should: be inexpensive to sample and evaluate while moving far enough to reduce autocorrelation without producing excessive rejection. These requirements trade off against one another, so proposal quality is judged by effective sample size per unit computation rather than acceptance rate alone.

1.11 Gibbs Sampling

1.11.1 Overview

The Gibbs sampler is one of the most widely used computational methods for Bayesian inference. It was first used in statistical physics and later popularized in statistics.

The method is particularly useful for high-dimensional target distributions.

1.11.2 Key Idea

The Gibbs sampler works by *sequentially sampling from univariate conditional distributions* rather than attempting to sample directly from a multivariate distribution.

1.11.3 Two-Stage Gibbs Sampler

Consider a bivariate posterior distribution

$$\pi(\theta, \lambda \mid x).$$

Algorithm

Initialization: Start with an arbitrary value $\lambda^{(0)}$. **Iteration t:** Sample

$$\theta^{(t)} \sim \pi_1(\theta \mid x, \lambda^{(t-1)}),$$

Sample

$$\lambda^{(t)} \sim \pi_2(\lambda \mid x, \theta^{(t)}).$$

The joint distribution $\pi(\theta, \lambda \mid x)$ is a stationary distribution for this Markov chain.

1.11.4 Multi-Stage Gibbs Sampler

General Form

For a joint distribution $\pi(\theta \mid x)$ with full conditionals $\pi_1, \ldots, \pi_p$, define the iterative scheme:

Given $(\theta_1^{(t)}, \ldots, \theta_p^{(t)})$,

$$\begin{aligned}
\theta_1^{(t+1)} &\sim \pi_1(\theta_1 \mid x, \theta_2^{(t)}, \ldots, \theta_p^{(t)}), \\
\theta_2^{(t+1)} &\sim \pi_2(\theta_2 \mid x, \theta_1^{(t+1)}, \theta_3^{(t)}, \ldots, \theta_p^{(t)}), \\
&\vdots \\
\theta_p^{(t+1)} &\sim \pi_p(\theta_p \mid x, \theta_1^{(t+1)}, \ldots, \theta_{p-1}^{(t+1)}).
\end{aligned}$$

Then

$$\theta^{(t)} \;\to\; \theta \sim \pi(\theta \mid x).$$

1.11.5 Remarks on Gibbs Sampling

Systematic-scan and random-scan Gibbs kernels both preserve the joint target, although they can have different convergence rates and neither universally dominates the other. Components need not be updated at every iteration, but each must be selected often enough to retain irreducibility. The full state sequence is Markov; an individual coordinate sequence generally is not, because its transition depends on unobserved current values of other blocks.

1.11.6 Convergence Diagnostics

Convergence is a property of a transition kernel as $t \to \infty$, not an event that can be certified at a finite iteration. Diagnostics search for evidence against approximate stationarity or adequate exploration. They cannot prove that an undiscovered mode does not exist.

Common Diagnostics

Run several chains from dispersed initial states. Rank-normalized split $\widehat{R}$ compares between-chain with within-chain variation after splitting each chain, while bulk and tail effective sample sizes quantify precision for central and tail summaries. Geweke compares early and late windows using spectral variance estimates; Heidelberger–Welch and Raftery–Lewis target stationarity and quantile precision. These procedures answer different questions and should be paired with trace plots and model-specific summaries.

Quick Checks of Convergence

A trace plot can reveal drift, sticking, and rare jumps. In mixture models, label switching may make component-specific traces look unstable even when the density is well explored; invariant functionals such as the predictive density or co-clustering matrix are more meaningful diagnostics. Conversely, several chains trapped in the same mode can look deceptively stable.

1.12 Autocorrelation

Mixing

The speed at which the chain explores the parameter space is called *mixing*.

High autocorrelation implies slow mixing and slow convergence.

Autocorrelation Function

For a sequence $\{\theta^{(1)}, \ldots, \theta^{(T)}\}$, the lag-$t$ autocorrelation function is

$$\mathrm{acf}_t(\theta) = \frac{\frac{1}{T-t}\sum_{j=1}^{T-t}(\theta^{(j)} - \bar{\theta})(\theta^{(j+t)} - \bar{\theta})}{\frac{1}{T-t}\sum_{j=1}^{T}(\theta^{(j)} - \bar{\theta})^2}.$$

Remarks

Gibbs correlation is determined by the blocking and parameterization, while a Metropolis proposal adds tuning choices. Adaptive schemes may learn scale and covariance during warmup, but unrestricted adaptation can destroy ergodicity.

Practical Implications

For a scalar functional h, the Markov-chain variance is governed by

$$\tau_h = 1 + 2\sum_{t\geq 1}\rho_t(h), \qquad \mathrm{ESS}_h = T/\tau_h.$$

Effective sample size is functional-specific. Thinning reduces file size but usually discards information; retaining correlated draws and estimating their spectral variance is statistically more efficient when storage is affordable.

1.13 Reading: Mixture Models and Bayesian Computation

Motivation for Mixtures

Mixture models provide a flexible framework for modelling complex data distributions:

$$f(x) = \sum_{i=1}^{k} p_i f(x \mid \theta_i), \quad \sum_{i=1}^{k} p_i = 1.$$

Mixtures serve both a structural and an approximation role. Structurally, a latent allocation represents unobserved subpopulations. As an approximation, a sufficiently rich collection of kernels can represent densities with skewness, multiple modes, or heavy tails even when no literal subpopulation interpretation is intended.

Thus, mixtures bridge parametric and nonparametric modeling.

Latent Variable Representation

Introduce allocation variables

$$Z_i \sim \text{Multinomial}(1; p_1, \ldots, p_k), \quad X_i \mid Z_i = j \sim f(x \mid \theta_j).$$

This transforms the model into a *complete-data formulation*, which is critical for computation.

Computational Challenge

The likelihood for n observations is

$$L(\theta, p \mid x) = \prod_{i=1}^{n} \sum_{j=1}^{k} p_j f(x_i \mid \theta_j).$$

Expanding this produces k^n terms, making direct inference intractable.

Implication: Closed-form Bayesian inference is generally impossible.

Combinatorial Explosion

Posterior inference can be written as

$$\pi(\theta, p \mid x) = \sum_{z} \omega(z)\, \pi(\theta, p \mid x, z),$$

where z are allocation vectors.

There are k^n allocation vectors, although only a small fraction usually carry appreciable posterior mass. Exact summation is therefore infeasible and motivates stochastic posterior exploration by MCMC.

EM vs Bayesian Computation

The EM algorithm alternates an E-step, which computes the conditional expected complete-data log likelihood, and an M-step, which maximizes that expectation. It is often efficient for maximum-likelihood estimation, but it is sensitive to initialization and converges only to a local stationary point. EM is thus an optimization procedure; MCMC instead attempts to represent the full posterior, including uncertainty across separated modes.

Mixtures as Ill-Posed Problems

Mixture inference is ill posed because some components can receive no data and, for common location–scale kernels, a component variance can collapse around one observation and make the likelihood unbounded. Small changes in the sample may consequently produce large parameter changes. Proper priors and posterior summaries invariant to label permutations regularize this inverse problem.

Label Switching and Identifiability

The likelihood is invariant under permutation:

$$(\theta_1, \ldots, \theta_k) \sim (\theta_{\sigma(1)}, \ldots, \theta_{\sigma(k)}).$$

With an exchangeable prior on component parameters, the posterior consequently has $k!$ symmetric modes. Component-labelled marginal means average across these modes and need not describe any fitted component. Density estimates, predictive distributions, and co-clustering probabilities are label-invariant; component-specific summaries require an explicit identifiability convention or post-processing rule.

Prior Design in Mixtures

Improper component priors are particularly dangerous because

$$\pi(\theta) = \prod_i \pi_i(\theta_i) \quad \Rightarrow \quad \text{the posterior may be improper.}$$

An empty component receives no likelihood information, so its conditional posterior remains its improper prior. Proper component priors avoid this failure; hierarchical dependence can additionally stabilize weakly populated components through partial pooling.

Mixtures and Nonparametric Bayes

Mixtures connect classical smoothing with Bayesian nonparametrics. A kernel density estimator,

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^{n} K(x - x_i)$$

is a finite mixture centered at the observations. A Dirichlet-process prior,

$$F \sim \mathrm{DP}(F_0, \alpha)$$

induces an infinite random mixture after a sampling kernel is introduced. Bernstein-polynomial mixtures such as

$$f(x) = \sum_k p_k \mathrm{Beta}(\alpha_k, \beta_k)$$

give another dense approximation family. Infinite mixtures are therefore one major BNP construction, although random functions and random partitions need not always be expressed as mixtures.

Key Takeaways for MCMC

Latent allocations turn an intractable mixture likelihood into tractable conditional updates, which Gibbs and Metropolis–Hastings samplers exploit. The resulting posterior remains multimodal and permutation-symmetric, so diagnostics and summaries must focus on inferentially meaningful, label-invariant quantities.

Chapter 2

Dirichlet Process

Bayesian Models Revisited

A Bayesian statistical model consists of:

A Bayesian model combines a likelihood $f(y \mid \theta)$ with a prior $\pi(\theta)$, so uncertainty about θ is represented probabilistically.

Uncertainty about parameters θ is modeled through the prior distribution.

Inference is based on the posterior distribution

$$\pi(\theta \mid y) = \frac{f(y \mid \theta)\pi(\theta)}{\int f(y \mid \theta)\pi(\theta)\, d\theta}.$$

2.1 Nonparametric Bayesian Priors

Modeling Idea

A *nonparametric Bayesian model* places a distribution on distributions.

$$\theta \mid G \sim G$$

where G itself is random:

$$G \sim P(G).$$

Uncertainty is therefore represented at two levels: individual latent values θ are random conditional on G, and the population distribution G is itself random. Depending on the application, inference may target latent values, the distribution G, or predictive functionals obtained from both.

2.2 Finite Mixture Model

Generative Model (Two Components)

For a two-component Gaussian mixture, let

$$z_n \overset{iid}{\sim} \text{Categorical}(\rho_1, \rho_2),$$

$$x_n \mid z_n \sim \mathcal{N}(\mu_{z_n}, \Sigma),$$

$$\mu_k \overset{iid}{\sim} \mathcal{N}(\mu_0, \Sigma_0),$$

and assign the mixing weight the prior

$$\rho_1 \sim \text{Beta}(a_1, a_2), \quad \rho_2 = 1 - \rho_1.$$

Inference Goal

Posterior inference jointly estimates the allocations z_n, the component parameters, and the mixing proportions; uncertainty in clustering is not discarded by fixing a single partition.

2.3 Beta Distribution Review

The Beta distribution is

$$\text{Beta}(\rho_1 \mid a_1, a_2) = \frac{\Gamma(a_1 + a_2)}{\Gamma(a_1)\Gamma(a_2)} \rho_1^{a_1 - 1}(1 - \rho_1)^{a_2 - 1}.$$

Gamma Function

The gamma function satisfies $\Gamma(x + 1) = x\Gamma(x)$ for $x > 0$ and reduces to $\Gamma(m + 1) = m!$ for nonnegative integers m.

Behavior

For a symmetric beta distribution, taking $a_1 = a_2$ toward zero concentrates mass near the boundaries, whereas large common values concentrate near $1/2$. When $a_1 > a_2$, the distribution shifts toward one.

Conjugacy

If

$$\rho_1 \sim \text{Beta}(a_1, a_2), \quad z \sim \text{Categorical}(\rho_1, \rho_2),$$

then

$$\rho_1 \mid z \sim \text{Beta}(a_1 + \mathbf{1}\{z = 1\},\ a_2 + \mathbf{1}\{z = 2\}).$$

2.4 Dirichlet Distribution

Definition

For $\rho_{1:K}$,

$$\text{Dirichlet}(\rho_{1:K} \mid a_{1:K}) = \frac{\Gamma\left(\sum_{k=1}^{K} a_k\right)}{\prod_{k=1}^{K} \Gamma(a_k)} \prod_{k=1}^{K} \rho_k^{a_k - 1},$$

with

$$\rho_k > 0, \quad \sum_{k=1}^{K} \rho_k = 1.$$

Construction via Gamma

Let

$$Z_j \sim \text{Gamma}(a_j, 1), \quad j = 1, \ldots, K,$$

and define

$$Y_j = \frac{Z_j}{\sum_{l=1}^{K} Z_l}.$$

Then

$$(Y_1, \ldots, Y_K) \sim \text{Dirichlet}(a_1, \ldots, a_K).$$

Special Case

For $K = 2$:

$$\text{Dirichlet}(a_1, a_2) = \text{Beta}(a_1, a_2).$$

Moments

$$E(Y_j) = \frac{a_j}{\sum_{l=1}^{K} a_l}, \quad E(Y_j^2) = \frac{a_j(a_j + 1)}{(\sum_l a_l)(1 + \sum_l a_l)}.$$

For $i \neq j$:

$$E(Y_i Y_j) = \frac{a_i a_j}{(\sum_l a_l)(1 + \sum_l a_l)}.$$

Marginals

$$Y_j \sim \text{Beta}\left(a_j, \sum_{i \neq j} a_i\right).$$

Conjugacy

If

$$\rho_{1:K} \sim \text{Dirichlet}(a_{1:K}), \quad z \sim \text{Categorical}(\rho_{1:K}),$$

then

$$\rho_{1:K} \mid z \sim \text{Dirichlet}(a'_1, \ldots, a'_K), \quad a'_k = a_k + \mathbf{1}\{z = k\}.$$

2.5 Finite Mixture Model (General K)

The general K-component model is

$$\rho_{1:K} \sim \text{Dirichlet}(a_{1:K}),$$

$$\mu_k \sim \mathcal{N}(\mu_0, \Sigma_0),$$

$$z_n \sim \text{Categorical}(\rho_{1:K}),$$

with observation model

$$x_n \mid z_n \sim \mathcal{N}(\mu_{z_n}, \Sigma).$$

2.6 When $K > N$

Interpretation

Here K is the number of available components, while the number of occupied clusters is the number represented by at least one observation. The latter is random, usually smaller than K, and tends to grow with sample size.

Key Observations

Choosing K can materially change the posterior and is especially awkward when data arrive sequentially. This motivates a prior with countably many available components but only finitely many occupied by any finite sample.

2.7 Choosing $K = \infty$

The limiting construction takes

$$K = \infty.$$

Infinite Mixing Weights

We seek a sequence of strictly positive weights

$$\rho = (\rho_1, \rho_2, \ldots) \quad \text{such that} \quad \sum_{k=1}^{\infty} \rho_k = 1.$$

Dirichlet Decomposition

From the finite Dirichlet:

$$\rho_{1:K} \sim \text{Dirichlet}(a_{1:K}),$$

we have the decomposition:

$$\rho_1 \sim \text{Beta}\Big(a_1,\ \sum_{k=2}^{K} a_k\Big),$$

and conditionally,

$$\frac{(\rho_2, \ldots, \rho_K)}{1-\rho_1} \sim \text{Dirichlet}(a_2, \ldots, a_K).$$

This suggests a recursive construction.

2.8 Stick-Breaking Construction

2.8.1 Finite Construction

Define

$$\begin{aligned} V_1 &\sim \text{Beta}(a_1, a_2 + a_3 + a_4), \quad \rho_1 = V_1, \\ V_2 &\sim \text{Beta}(a_2, a_3 + a_4), \quad \rho_2 = (1 - V_1)V_2, \\ V_3 &\sim \text{Beta}(a_3, a_4), \quad \rho_3 = (1 - V_1)(1 - V_2)V_3, \end{aligned}$$

and so on, with

$$\rho_K = 1 - \sum_{k=1}^{K-1} \rho_k.$$

This is the *stick-breaking* representation.

2.8.2 Infinite Stick-Breaking

Taking $K \to \infty$, define

$$V_k \sim \text{Beta}(a_k, b_k),$$

and construct weights:

$$\rho_k = \left(\prod_{j=1}^{k-1}(1 - V_j)\right) V_k.$$

2.8.3 GEM Distribution

A special case of stick-breaking is:

$$a_k = 1, \quad b_k = \alpha > 0.$$

Then

$$\rho = (\rho_1, \rho_2, \ldots) \sim \mathrm{GEM}(\alpha),$$

with

$$V_k \sim \mathrm{Beta}(1, \alpha), \quad \rho_k = \left(\prod_{j=1}^{k-1} (1 - V_j) \right) V_k.$$

This defines a random probability distribution over a countably infinite set.

2.9 Interpretation

Hierarchy of Distributions

The beta distribution randomizes probabilities on two categories, the Dirichlet distribution extends this to a fixed finite number, and GEM stick-breaking produces random probabilities on countably many categories.

Key Properties

Although infinitely many components are available, a finite sample occupies only finitely many of them. The number occupied grows with N, making the effective dimension data-adaptive.

Conceptual Insight

The formal parameter contains infinitely many component weights and atoms, but finite-sample computation depends primarily on the occupied clusters and a controlled representation of the remaining tail.

2.10 Readings: Foundations of the Dirichlet Process

Dirichlet Process as a Distribution over Distributions

A Dirichlet process defines a *random probability measure* P on a measurable space $(\mathcal{X}, \mathcal{B})$.

Thus P itself is the parameter and takes values in a space of probability measures. Its defining finite-dimensional property is that, for every measurable partition $(B_1, \ldots, B_k)$,

For any measurable partition $(B_1, \ldots, B_k)$,

$$\Big(P(B_1), \ldots, P(B_k)\Big) \sim \text{Dirichlet}\Big(a(B_1), \ldots, a(B_k)\Big),$$

where a is a finite base measure.

Interpretation of the Base Measure

Let

$$a(\mathcal{X}) = \alpha, \quad \beta(\cdot) = \frac{a(\cdot)}{\alpha}.$$

The normalized measure β is the base distribution and α is the total concentration. In particular,

$$E[P(B)] = \beta(B)$$

Thus, the Dirichlet process is centered around β.

Discrete Nature of the Dirichlet Process

A draw from a DP is discrete almost surely:

$$P = \sum_{n=1}^{\infty} p_n \delta_{Y_n}$$

even if β is continuous.

Consequently independent latent draws from P can repeat exactly, inducing a random partition even when the base distribution β is continuous.

Posterior Conjugacy

If

$$P \sim \text{DP}(a), \quad X \mid P \sim P,$$

then the posterior is:

$$P \mid X \sim \text{DP}(a + \delta_X).$$

For data $(X_1, \ldots, X_n)$:

$$P \mid X_{1:n} \sim \text{DP}\left(a + \sum_{i=1}^{n} \delta_{X_i}\right).$$

This is the infinite-dimensional analogue of Dirichlet conjugacy.

Constructive Representation (Stick-Breaking)

A constructive representation is

$$P = \sum_{n=1}^{\infty} p_n \delta_{Y_n}$$

where $Y_n \overset{iid}{\sim} \beta$ and the weights are generated by

$$p_1 = V_1, \quad p_n = V_n \prod_{j=1}^{n-1} (1 - V_j)$$

$$V_n \sim \text{Beta}(1, \alpha)$$

This gives an explicit generative construction of the random measure.

Distributional Fixed-Point Property

The Dirichlet process satisfies the distributional equation:

$$P = \theta \delta_Y + (1 - \theta) P',$$

where $\theta \sim \text{Beta}(1, \alpha)$, $Y \sim \beta$, and $P' \overset{d}{=} P$ are mutually independent. The identity isolates the first size-biased atom and leaves a rescaled independent DP draw in the remainder.

This recursive structure underlies stick-breaking.

Connection to Bayesian Nonparametrics

The DP can model an unknown distribution directly or serve as a mixing measure in an infinite mixture. It replaces a fixed component count by

$$K = \infty \quad \text{but effective clusters } < \infty.$$

Complexity is therefore controlled through the concentration parameter and the posterior partition rather than by preselecting K.

Practical Insight

For finite data, posterior algorithms need represent only finitely many occupied atoms plus an aggregate or truncated tail. Conjugacy and predictive representations keep the updates tractable, so

$$\text{Unknown number of clusters} \Rightarrow \text{handled probabilistically.}$$

Distributions Hierarchy Interpretation

The hierarchy progresses from beta probabilities on two categories, through Dirichlet probabilities on K categories, to GEM weights on countably many indices. Attaching independent atoms in a general space Φ converts those weights into a Dirichlet-process random measure:

Formally, the Dirichlet process admits the representation:

$$\rho = (\rho_1, \rho_2, \dots) \sim \text{GEM}(\alpha), \quad \phi_k \overset{i.i.d.}{\sim} G_0,$$

$$G = \sum_{k=1}^{\infty} \rho_k \delta_{\phi_k}.$$

Here ρ_k is the mass of cluster k and ϕ_k is its location or component parameter. G is a discrete random measure.

2.10.1 Dirichlet Process Mixture Model (DPMM)

Gaussian mixture case

The weights satisfy

$$\rho \sim \text{GEM}(\alpha)$$

The component parameters satisfy

$$\mu_k \overset{i.i.d.}{\sim} \mathcal{N}(\mu_0, \Sigma_0)$$

The random measure is

$$G = \sum_{k=1}^{\infty} \rho_k \delta_{\mu_k} \equiv DP(\alpha, \mathcal{N}(\mu_0, \Sigma_0))$$

The latent variables follow

$$z_n \overset{i.i.d.}{\sim} \text{Categorical}(\rho), \quad \mu_n^* = \mu_{z_n}, \quad \mu_n^* \overset{i.i.d.}{\sim} G$$

Conditional on the latent variables, the observations follow

$$x_n \mid \mu_n^* \sim \mathcal{N}(\mu_n^*, \Sigma)$$

General formulation

More generally:

$$\rho \sim \text{GEM}(\alpha), \quad \phi_k \overset{i.i.d.}{\sim} G_0$$

$$G = \sum_{k=1}^{\infty} \rho_k \delta_{\phi_k} \equiv DP(\alpha, G_0)$$

$$z_n \sim \text{Categorical}(\rho), \quad \theta_n = \phi_{z_n}, \quad \theta_n \overset{i.i.d.}{\sim} G$$

$$x_n \mid \theta_n \sim F(\theta_n)$$

Dirichlet Process Definition

Let $(\mathcal{Y}, \mathcal{B}(\mathcal{Y}))$ be a measurable space.

Theorem 2.10.1. *(Dirichlet Process) A random probability measure F satisfies*

$$F \sim DP(\alpha, H)$$

if for any measurable partition $(B_1, \ldots, B_n)$,

$$(F(B_1), \ldots, F(B_n)) \sim \text{Dirichlet}(\alpha H(B_1), \ldots, \alpha H(B_n)).$$

The family of Dirichlet finite-dimensional distributions is projectively consistent under aggregation of partition cells. Kolmogorov consistency therefore defines a random probability measure F, rather than merely a separate Dirichlet vector for each chosen partition.

Properties of the Dirichlet Process

Moments

For any measurable A,

$$E[F(A)] = H(A), \quad \text{Var}(F(A)) = \frac{H(A)(1 - H(A))}{1 + \alpha}.$$

Self-similarity

For B with $H(B) > 0$, normalize the restriction by $F_B(A) = F(A \cap B)/F(B)$ and $H_B(A) = H(A \cap B)/H(B)$. Then

$$F_B \sim DP\{\alpha H(B), H_B\},$$

independently of the total mass $F(B)$ conditional on the corresponding Dirichlet decomposition. The concentration changes from α to $\alpha H(B)$; omitting this factor overstates local prior information.

Conjugacy

If

$$y_i \overset{i.i.d.}{\sim} F, \quad F \sim DP(\alpha, H),$$

then

$$F \mid y \sim DP\left(\alpha + n,\ \frac{1}{\alpha + n} \sum_{i=1}^{n} \delta_{y_i} + \frac{\alpha}{\alpha + n} H\right).$$

In particular, the posterior mean is

$$E[F(A) \mid y] = \frac{1}{\alpha + n} \sum_{i=1}^{n} \delta_{y_i}(A) + \frac{\alpha}{\alpha + n} H(A).$$

Proof. Fix a measurable partition $(A_1, \ldots, A_r)$. The prior vector of masses is Dirichlet with parameters $(\alpha H(A_1), \ldots, \alpha H(A_r))$. Conditional on G, the cell counts are multinomial. Dirichlet–multinomial conjugacy therefore changes the parameters to $\alpha H(A_j) + \sum_i \mathbf{1}\{y_i \in A_j\}$. Because this holds for every finite measurable partition and the resulting laws are consistent, they define the stated posterior Dirichlet process. □

The posterior base measure is a convex combination of H and the empirical measure. The concentration changes from α to $\alpha + n$, so posterior random measures fluctuate less around their posterior mean as data accrue.

Richness

A draw from a DP is almost surely discrete even when H has a density. This does not mean that its weak support contains only discrete target laws. If the support of H is the whole sample space, the DP has full weak support on probability measures on that space: every weak neighborhood of a target probability measure receives positive prior probability. Density estimation usually places the DP on a mixing distribution and convolves its discrete atoms with a continuous kernel; the resulting mixture density is continuous.

Support alone is not a contraction-rate theorem. Posterior consistency also requires the true distribution to lie in an appropriate Kullback–Leibler support and the model to admit tests separating neighborhoods of the truth.

Stick-Breaking Theorem

Theorem 2.10.2. *(Sethuraman) Let*

$$\theta_k \overset{i.i.d.}{\sim} H, \quad V_k \overset{i.i.d.}{\sim} \mathrm{Beta}(1, \alpha),$$

and define

$$\pi_k = V_k \prod_{l=1}^{k-1} (1 - V_l).$$

Then

$$\sum_{k=1}^{\infty} \pi_k \delta_{\theta_k} \sim DP(\alpha, H).$$

Posterior Stick-Breaking Interpretation

Given

$$G \sim DP(\alpha, H), \quad \theta \sim G,$$

we obtain:

$$G \mid \theta \sim DP\left(\alpha + 1,\ \frac{\alpha H + \delta_\theta}{\alpha + 1}\right).$$

This implies a decomposition:

$$G = V\delta_\theta + (1 - V)G', \quad V \sim \text{Beta}(1, \alpha),$$

where G' is the residual measure.

Recursive Property of the DP

For a partition $(\theta, A_1, \ldots, A_K)$:

$$(G(\theta), G(A_1), \ldots, G(A_K)) \sim \text{Dirichlet}(1, \alpha H(A_1), \ldots, \alpha H(A_K)).$$

This implies:

$$G' \sim DP(\alpha, H),$$

showing the *self-similar recursive structure* of the Dirichlet process.

Key Insight

The finite-dimensional definition, stick-breaking construction, and Blackwell–MacQueen predictive rule describe the same law from different directions. The first establishes a random probability measure, the second exhibits its almost-sure discreteness, and the third exposes the induced random partition. A finite dataset occupies only finitely many atoms even though the random measure contains countably many, which makes infinite mixture models computationally accessible.

Stick-Breaking Expansion

We can recursively expand the Dirichlet process draw:

$$G \sim DP(\alpha, H)$$

$$G = V_1\delta_{\theta_1^*} + (1 - V_1)G_1$$

$$G = V_1\delta_{\theta_1^*} + (1 - V_1)\Big(V_2\delta_{\theta_2^*} + (1 - V_2)G_2\Big)$$

Continuing:

$$G = \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k^*},$$

where

$$\pi_k = V_k \prod_{l=1}^{k-1}(1 - V_l), \quad V_k \sim \text{Beta}(1, \alpha), \quad \theta_k^* \sim H.$$

Interpretation

Draws from a DP are discrete measures. Stick-breaking allocates their masses, while IID draws from H determine atom locations. If H is nonatomic, the atoms are distinct almost surely; ties among observations arise because they select the same random atom, not because H itself produces ties.

Marginalizing the Weights

Consider a simple case:

$$\rho_1 \sim \text{Beta}(a_1, a_2), \quad z_n \sim \text{Categorical}(\rho_1, \rho_2).$$

Integrating out ρ_1, we obtain:

$$p(z_n = 1 \mid z_{1:n-1}) = \frac{a_{1,n}}{a_{1,n} + a_{2,n}},$$

where

$$a_{1,n} = a_1 + \sum_{m=1}^{n-1} \mathbf{1}\{z_m = 1\}, \quad a_{2,n} = a_2 + \sum_{m=1}^{n-1} \mathbf{1}\{z_m = 2\}.$$

2.10.2 Pólya Urn Interpretation

This predictive rule corresponds to a Pólya urn:

Choose a ball with probability proportional to its count; Replace it and add another ball of the same color.

Then

$$\lim_{n\to\infty} \frac{\#\text{ orange}}{\#\text{ total}} \overset{d}{=} \text{Beta}(a_{\text{orange}}, a_{\text{green}}).$$

Multivariate Extension

For

$$\rho_{1:K} \sim \text{Dirichlet}(a_{1:K}), \quad z_n \sim \text{Categorical}(\rho_{1:K}),$$

we obtain:

$$p(z_n = k \mid z_{1:n-1}) = \frac{a_{k,n}}{\sum_{j=1}^{K} a_{j,n}},$$

where

$$a_{k,n} = a_k + \sum_{m=1}^{n-1} \mathbf{1}\{z_m = k\}.$$

This corresponds to a *multivariate Pólya urn.*

2.10.3 Blackwell–MacQueen Pólya Urn

We now consider the infinite limit.

Let:

$$G \sim DP(\alpha, H), \quad \theta_i \mid G \sim G.$$

Then

First sample

$$\theta_1 \sim H, \quad G \mid \theta_1 \sim DP\left(\alpha + 1, \frac{\alpha H + \delta_{\theta_1}}{\alpha + 1}\right).$$

Second sample

$$\theta_2 \mid \theta_1 \sim \frac{\alpha H + \delta_{\theta_1}}{\alpha + 1}.$$

General step

$$\theta_n \mid \theta_{1:n-1} \sim \frac{\alpha H + \sum_{i=1}^{n-1} \delta_{\theta_i}}{\alpha + n - 1}.$$

Urn Interpretation

With probability proportional to α, draw a new value from H; With probability proportional to existing counts, reuse a previous value.

Equivalently:

Draw a ball proportional to its mass; If it is *new*, create a new color; Otherwise, reinforce the existing color.

Key Insight

This produces a sequence:

$$\theta_1, \theta_2, \ldots$$

with: clustering (shared values) increasing number of clusters; probability of new cluster controlled by α.

2.11 Posterior Support, Consistency, and Contraction

Large topological support is only a prior property. Posterior consistency asks whether the posterior concentrates near the data-generating distribution as the sample size grows, while a contraction rate quantifies how quickly that concentration occurs in a chosen metric.

Definition 2.11.1 (Kullback–Leibler support)**.** A prior Π on densities has f_0 in its KL support if, for every $\varepsilon > 0$,

$$\Pi\left(f : \int f_0 \log \frac{f_0}{f} \, \mathrm{d}\mu < \varepsilon\right) > 0.$$

Theorem 2.11.1 (Schwartz's consistency theorem)**.** *Let $X_1, X_2, \ldots$ be IID from f_0. Suppose f_0 lies in the KL support of Π. If for every neighborhood U of f_0 there exist tests ϕ_n such that*

$$P_{f_0}^n \phi_n \to 0, \qquad \sup_{f \in U^c} P_f^n (1 - \phi_n) \to 0,$$

then $\Pi(U^c \mid X_{1:n}) \to 0$ in P_{f_0}-probability.

Proof idea. Write posterior mass as a numerator over the marginal-likelihood denominator. KL prior mass prevents the denominator from becoming exponentially too small. The tests split the numerator over U^c into a term controlled under f_0 and a uniformly small type-II-error term under alternatives. Markov's inequality then yields posterior concentration. The theorem separates the two essential ingredients: local prior thickness near the truth and statistical distinguishability of its complement. □

For a rate $\varepsilon_n \downarrow 0$, a common contraction theorem requires three ingredients on a sieve $\mathcal{F}_n$:

$$\log N(\varepsilon_n, \mathcal{F}_n, d) \lesssim n\varepsilon_n^2,$$

$$\Pi(\mathcal{F}_n^c) \le e^{-cn\varepsilon_n^2}, \qquad \Pi\{\text{KL neighborhood of } f_0\} \ge e^{-Cn\varepsilon_n^2}.$$

The entropy bound controls the number of tests needed, the sieve-tail bound prevents excessive prior mass on overly complex regions, and the small-ball condition ensures sufficient mass near f_0. Under metric-specific testing conditions these imply

$$\Pi\{f : d(f, f_0) > M\varepsilon_n \mid X_{1:n}\} \to 0$$

for sufficiently large M.

Example 2.11.1 (DP Gaussian mixtures)**.** A DP prior is placed on a mixing distribution G, while the observable density is

$$f_{G,\sigma}(x) = \int \phi_\sigma(x - \mu) G(\, \mathrm{d}\mu).$$

Although G is discrete almost surely, $f_{G,\sigma}$ is smooth. Consistency requires the base measure to put mass near the locations needed to approximate f_0 and the scale prior to put mass near suitable bandwidths. Contraction rates balance Gaussian-mixture approximation error against the prior cost of using many atoms and small scales. Thus discreteness of G is not an obstacle to continuous density estimation.

Weak consistency, Hellinger consistency, and consistency of the number of clusters are different claims. A posterior can estimate the mixture density consistently while continuing to create small redundant clusters. Any theorem or empirical summary should state the target object and topology explicitly.

2.12 Chinese Restaurant Process (CRP)

Construction

The Chinese Restaurant Process provides an equivalent formulation of the Dirichlet process clustering mechanism.

Each observation (customer) arrives sequentially. Customer n: sits at an existing table k with probability proportional to the number of customers already there; starts a new table with probability proportional to α.

Formally:

$$p(z_n = k \mid z_{1:n-1}) = \begin{cases} \frac{n_k}{n-1+\alpha}, & \text{existing table } k, \\ \frac{\alpha}{n-1+\alpha}, & \text{new table.} \end{cases}$$

Partition Interpretation

Let Π_N denote the partition of $\{1, \ldots, N\}$ induced by assignments $z_1, \ldots, z_N$.

Example:

$$z_1 = z_2 = z_7 = z_8 = 1, \quad z_3 = z_5 = z_6 = 2, \quad z_4 = 3$$
$$\Rightarrow \Pi_8 = \{\{1, 2, 7, 8\}, \{3, 5, 6\}, \{4\}\}.$$

A partition consists of mutually exclusive and exhaustive subsets of $\{1, \ldots, N\}$.

Probability of a Seating Arrangement

Let K_N be the number of tables and let $C \in \Pi_N$ denote a cluster.

Then

$$\mathbb{P}(\Pi_N = \pi_N) = \frac{\alpha^{K_N} \prod_{C \in \Pi_N} (|C| - 1)!}{\alpha(\alpha + 1) \cdots (\alpha + N - 1)}.$$

This is the exchangeable partition probability function (EPPF). To derive it, multiply the sequential CRP probabilities. Every new table after the first contributes a factor α, a table that ends with size n_k contributes $1 \cdot 2 \cdots (n_k - 1) = (n_k - 1)!$, and the denominators multiply to the rising factorial $\alpha^{(N)} = \alpha(\alpha + 1) \cdots (\alpha + N - 1)$. The result depends only on block sizes, proving exchangeability of the induced partition.

Example 2.12.1. For $N = 4$, compare partitions with sizes $(3, 1)$ and $(2, 2)$. Their probabilities for a particular labeled partition are proportional to

$$\alpha^2(3-1)!(1-1)! = 2\alpha^2, \qquad \alpha^2(2-1)!^2 = \alpha^2.$$

The rich-get-richer mechanism favors the more imbalanced block sizes, although there are different numbers of labeled partitions of each type.

Exchangeability

The probability of a partition does not depend on the order of customers:

$$\mathbb{P}(\Pi_N = \pi_N) \text{ is invariant under permutations.}$$

Thus, the CRP is *exchangeable*.

Conditional Assignment Rule

Given $\Pi_{N,-n}$ (partition without customer n):

$$p(\Pi_N \mid \Pi_{N,-n}) = \begin{cases} \frac{|C|}{\alpha+N-1}, & \text{if } n \text{ joins cluster } C, \\ \frac{\alpha}{\alpha+N-1}, & \text{if } n \text{ creates new cluster.} \end{cases}$$

Representations of the Dirichlet Process

The Dirichlet process admits several equivalent representations:

Posterior form:

$$G \sim DP(\alpha, H), \quad \theta \mid G \sim G$$

$$\iff \theta \sim H, \quad G \mid \theta \sim DP\left(\alpha + 1, \frac{\alpha H + \delta_\theta}{\alpha + 1}\right)$$

Pólya urn:

$$\theta_n \mid \theta_{1:n-1} \sim \frac{\alpha H + \sum_{i=1}^{n-1} \delta_{\theta_i}}{\alpha + n - 1}$$

CRP:

$$p(z_n = k \mid z_{1:n-1}) = \begin{cases} \frac{n_k}{n-1+\alpha}, \\ \frac{\alpha}{n-1+\alpha} \end{cases}$$

Stick-breaking:

$$\pi_k = \beta_k \prod_{i=1}^{k-1}(1-\beta_i), \quad \beta_k \sim \text{Beta}(1, \alpha)$$

$$G = \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k^*}$$

Exchangeability and Joint Distribution

From the predictive rule:

$$\theta_n \mid \theta_{1:n-1} \sim \frac{\alpha H + \sum_{i=1}^{n-1} \delta_{\theta_i}}{\alpha + n - 1},$$

the joint distribution is:

$$p(\theta_1, \ldots, \theta_n) = \frac{\alpha^K \prod_{k=1}^{K} h(\theta_k^*)(m_k - 1)!}{\prod_{i=1}^{n}(\alpha + i - 1)},$$

where: θ_k^* are distinct values; m_k are their multiplicities; $h(\theta)$ is the density under H.

This distribution is exchangeable.

By De Finetti's theorem, this implies the existence of a random measure G such that

$$\theta_1, \theta_2, \ldots \overset{iid}{\sim} G, \quad G \sim DP(\alpha, H).$$

2.13 Number of Clusters

Let K_n denote the number of clusters among n observations.

2.13.1 Expected Value

Define indicator:

$$I_i = \mathbf{1}\{\text{new cluster at step } i\}.$$

Then

$$\mathbb{E}[K_n] = \sum_{i=1}^{n} \mathbb{E}[I_i] = \sum_{i=1}^{n} \frac{\alpha}{\alpha + i - 1}.$$

Therefore,

$$\mathbb{E}[K_n] = O(\alpha \log n).$$

More precisely, using the digamma function ψ,

$$\mathbb{E}(K_n) = \alpha\{\psi(\alpha + n) - \psi(\alpha)\} = \alpha \log n + O(1).$$

The indicators I_i are independent Bernoulli variables under the sequential construction because the conditional probability of a new table at step i is the deterministic number $\alpha/(\alpha + i - 1)$. This observation yields both the variance formula below and a central limit theorem for K_n after centering and scaling.

2.13.2 Variance

$$\mathrm{Var}(K_n) = \sum_{i=1}^{n} \frac{\alpha(i-1)}{(\alpha + i - 1)^2}.$$

2.13.3 Distribution of K_n

The distribution satisfies:

$$\mathbb{P}(K_n = k) = \frac{\alpha}{\alpha + n - 1}\mathbb{P}(K_{n-1} = k - 1) + \frac{n - 1}{\alpha + n - 1}\mathbb{P}(K_{n-1} = k).$$

Boundary conditions:

$$\mathbb{P}(K_1 = 1) = 1.$$

Closed form:

$$\mathbb{P}(K_n = k) = C_n(k)\,\alpha^k \frac{\Gamma(\alpha)}{\Gamma(\alpha + n)},$$

where $C_n(k)$ are unsigned Stirling numbers of the first kind.

2.13.4 Interpretation

The DP partition has a rich-get-richer allocation rule and logarithmic expected cluster growth. This is a prior statement about latent partition complexity, not a guarantee that the posterior consistently estimates a finite true number of mixture components. In fact, a DP mixture can consistently estimate a density while its posterior on the number of occupied clusters remains inconsistent for the number of components in a finite mixture. Density estimation and component counting are distinct inferential goals.

Chapter 3

Sampling Techniques

3.1 Sampling Techniques for DP Mixture Models

Gaussian Likelihood Setup

Assume:

$$y \sim \mathcal{N}(\mu, \sigma^2), \quad p(y \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y-\mu)^2\right).$$

For i.i.d. data $y = (y_1, \ldots, y_n)$:

$$L(\mu, \sigma^2) \propto \left(\frac{1}{\sigma^2}\right)^{n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_i (y_i - \mu)^2\right).$$

We focus on:

$$p(\mu, \sigma^2 \mid y) = p(\mu \mid \sigma^2, y)\, p(\sigma^2 \mid y).$$

Semi-Conjugate Prior

Assume independent priors:

$$\pi(\mu, \sigma^2) = \pi(\mu)\pi(\sigma^2),$$

$$\mu \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad \sigma^2 \sim IG\left(\frac{\nu_0}{2}, \frac{\delta_0}{2}\right).$$

Then

$$\mu \mid \sigma^2, y \sim \mathcal{N}(\mu_n, \tau_n^2),$$

$$\mu_n = \frac{\mu_0/\sigma_0^2 + \bar{y}\, n/\sigma^2}{1/\sigma_0^2 + n/\sigma^2}, \quad \tau_n^2 = \frac{1}{1/\sigma_0^2 + n/\sigma^2}.$$

$$\sigma^2 \mid y \text{ not in closed form}, \quad \sigma^2 \mid y, \mu \sim IG.$$

3.1.1 Finite Mixture Models

A flexible density:

$$f(y) = \sum_{k=1}^{K} \tau_k f(y \mid \theta_k),$$

where:

$$\tau_k \in (0,1), \quad \sum_k \tau_k = 1.$$

3.1.2 Bayesian Mixture Model

$$y_i \mid \tau, \theta \sim \sum_{k=1}^{K} \tau_k f(y_i \mid \theta_k),$$

$$\tau \sim \mathrm{Dir}(\alpha, \ldots, \alpha), \quad \theta_k \sim \pi_k(\theta_k), \quad K \sim \pi(K).$$

3.1.3 Gibbs Sampling (Finite Case)

For fixed K:

The allocation distribution is

$$P(z_i^{(t)} = k \mid \cdot) \propto \tau_k^{(t-1)} f(y_i \mid \theta_k^{(t-1)}).$$

The weights satisfy

$$\tau \mid z \sim \mathrm{Dir}(\alpha_1 + \hat{n}_1, \ldots, \alpha_K + \hat{n}_K),$$

where $\hat{n}_k = \sum_i \mathbf{1}(z_i = k)$. The component parameters follow

$$p(\theta_k \mid \cdot) \propto \pi_k(\theta_k) \prod_{i:z_i=k} f(y_i \mid \theta_k).$$

3.2 CRP Mixture Model

Generative Model

$$\Pi_N \sim CRP(N, \alpha),$$

$$\forall C \in \Pi_N, \quad \phi_C \overset{i.i.d.}{\sim} G_0,$$

$$x_n \mid \phi_C \sim F(\phi_C), \quad n \in C.$$

Inference Goal

$$p(\Pi_N \mid x_{1:N}).$$

Gibbs Sampler (Collapsed)

$$p(\Pi_N \mid \Pi_{N,-n}, x) = \begin{cases} \frac{|C|}{\alpha+N-1}\, p(x_{C\cup\{n\}} \mid x_C), & \text{join } C, \\ \frac{\alpha}{\alpha+N-1}\, p(x_n), & \text{new cluster.} \end{cases}$$

For Gaussian case:

$$p(x_{C\cup\{n\}} \mid x_C) = \mathcal{N}(\tilde{m}, \tilde{\Sigma} + \Sigma),$$

$$\tilde{\Sigma}^{-1} = \Sigma_0^{-1} + |C|\Sigma^{-1}, \quad \tilde{m} = \tilde{\Sigma}\left(\Sigma^{-1}\sum_{m\in C} x_m + \Sigma_0^{-1}\mu_0\right).$$

3.3 DP Mixture Model (DPM)

$$y_i \sim \int p(y_i \mid \theta) G(d\theta), \quad G \sim DP(M, G_0).$$

Example:

$$y_i \sim \int \mathcal{N}(y_i \mid \mu, \sigma^2) G(d\mu).$$

3.4 Alternative Representations

3.4.1 Stick-Breaking Form

$$y_i \mid (w_h, \tilde{\theta}_h) \sim \sum_{h=1}^{\infty} w_h p(y_i \mid \tilde{\theta}_h),$$

$$\tilde{\theta}_h \sim G_0, \quad w_h = v_h \prod_{k<h}(1 - v_k), \quad v_h \sim \text{Beta}(1, M).$$

3.4.2 Clustering Representation

$$y_i \mid \theta_i \sim p(y_i \mid \theta_i), \quad \theta_i \mid G \sim G, \quad G \sim DP(M, G_0).$$

Integrating out G:

$$(\theta_1, \ldots, \theta_n) \sim \text{CRP-induced distribution.}$$

The predictive distribution is

$$p(\theta_{n+1} \mid \theta_{1:n}) \propto \sum_{j=1}^{k_n} n_j \delta_{\theta_j^*} + MG_0.$$

3.5 Collapsed Gibbs Sampling

3.5.1 Collapsed assignment probabilities

Define the prior and cluster-specific predictive densities

$$m_0(y) = \int p(y \mid \theta) G_0(\,\mathrm{d}\theta), \qquad m_C(y) = \int p(y \mid \theta) p(\theta \mid y_C)\,\mathrm{d}\theta.$$

After removing observation i, Bayes' rule and the CRP predictive law give

$$p(s_i = j \mid s_{-i}, y) \propto \begin{cases} n_{-i,j} m_{C_j}(y_i), & j = 1, \ldots, K_{-i}, \\ M m_0(y_i), & j = \text{new}. \end{cases}$$

The cluster size is the partition-prior contribution and the predictive density measures compatibility with the data already assigned to that cluster. Using only sizes would sample from the prior CRP, not the posterior.

Algorithm 1 One collapsed Gibbs sweep for a conjugate DPM

1: **for** $i = 1, \ldots, n$ **do**
2: Remove i and delete its old cluster if it becomes empty.
3: Evaluate $n_{-i,j} m_{C_j}(y_i)$ for occupied clusters.
4: Evaluate the new-cluster weight $M m_0(y_i)$.
5: Normalize these weights and sample the new assignment s_i.
6: Update the selected cluster's sufficient statistics.
7: **end for**

When conjugacy makes the predictive integrals analytic, component parameters need not be represented in the assignment chain. Rao–Blackwellization often reduces Monte Carlo variance and improves mixing.

3.5.2 Cluster Parameter Update

$$p(\theta_j^* \mid s, y) \propto G_0(\theta_j^*) \prod_{i : s_i = j} p(y_i \mid \theta_j^*).$$

3.6 Gaussian DPM Example

$$p(y_i \mid \theta_i) = \mathcal{N}(\theta_i, \sigma^2), \quad G_0 = \mathcal{N}(m, B).$$

$$\int p(y_i \mid \theta) dG_0(\theta) = \mathcal{N}(y_i \mid m, B + \sigma^2).$$

Cluster predictive:

$$\int p(y_i \mid \theta_j^*) dp(\theta_j^* \mid y_j^-) = \mathcal{N}(y_i \mid m_j^-, V_j^- + \sigma^2),$$

$$V_j^- = \left(\frac{1}{B} + \frac{n_j^-}{\sigma^2}\right)^{-1}, \quad m_j^- = V_j^- \left(\frac{m}{B} + \frac{1}{\sigma^2}\sum_{h \in S_j^-} y_h\right).$$

The posterior is

$$p(\theta_j^* \mid y) = \mathcal{N}(m_j, V_j).$$

3.7 Reading: Mixture of Dirichlet Process Models

Model Representation

The Dirichlet process mixture model is

$$y_i \mid \theta_i \sim F(\theta_i), \quad \theta_i \sim G, \quad G \sim DP(\alpha, G_0)$$

Integrating out G gives

$$\theta_i \mid \theta_{1:i-1} = \sum_{j<i} \frac{1}{i-1+\alpha}\delta_{\theta_j} + \frac{\alpha}{i-1+\alpha}G_0$$

This induces clustering through repeated values of θ_i.

Partition View

Let c_i denote cluster assignments. Then

$$P(c_i = c \mid c_{-i}) = \begin{cases} \frac{n_{-i,c}}{n-1+\alpha} \\ \frac{\alpha}{n-1+\alpha} \end{cases}$$

and

$$y_i \mid c_i = c \sim F(\theta_c)$$

Thus inference reduces to learning

$$(\Pi_n, \{\theta_c\})$$

Non-Conjugate Difficulty

Key terms required in Gibbs sampling are

$$\int F(y_i \mid \theta)\, dG_0(\theta) \quad \text{and} \quad p(\theta \mid y_i) \propto F(y_i \mid \theta)G_0(\theta)$$

These are not available in closed form in general.

No-Gaps Representation

Clusters are indexed as $1, \ldots, k$ with no empty labels. This avoids redundant representations of the same partition and simplifies computation.

Augmented Sampling Idea

Introduce auxiliary parameters

$$\theta_1^*, \ldots, \theta_m^* \sim G_0$$

Then approximate the new cluster term via

$$\alpha \int F(y_i \mid \theta)\, dG_0(\theta) \approx \frac{\alpha}{m} \sum_{j=1}^{m} F(y_i \mid \theta_j^*)$$

Gibbs Updates

Cluster assignments are updated via

$$P(c_i = c \mid \cdot) \propto \begin{cases} n_{-i,c} F(y_i \mid \theta_c) \\ \frac{\alpha}{m} F(y_i \mid \theta_c^*) \end{cases}$$

Cluster parameters are updated via

$$\theta_c \mid y_c \sim p(\theta_c \mid y_c)$$

Main Insight

Non-conjugacy can be handled by expanding the state space and performing Gibbs updates in the augmented model, avoiding direct evaluation of intractable integrals.

3.8 Reading: Sampling Methods for Dirichlet Process Mixtures

Dirichlet Process Mixture Representation

The model is

$$y_i \mid \theta_i \sim F(\theta_i), \quad \theta_i \sim G, \quad G \sim DP(\alpha, G_0)$$

which implies

$$\theta_i \mid \theta_{1:i-1} = \sum_{j<i} \frac{1}{i-1+\alpha} \delta_{\theta_j} + \frac{\alpha}{i-1+\alpha} G_0$$

This is equivalent to a Chinese restaurant process on cluster assignments.

Collapsed Gibbs Sampling

If conjugate, integrate out θ_c:

$$P(c_i = c \mid c_{-i}, y) \propto n_{-i,c} \cdot p(y_i \mid y_{c,-i})$$

For a new cluster:

$$\propto \alpha \int F(y_i \mid \theta)\, dG_0(\theta)$$

Failure Under Non-Conjugacy

The integral

$$\int F(y_i \mid \theta)\, dG_0(\theta)$$

is not tractable, so collapsed Gibbs cannot be used.

Metropolis-Hastings Update

Propose

$$\theta^* \sim G_0$$

Accept with probability

$$\rho = \min\left(1, \frac{F(y_i \mid \theta^*)}{F(y_i \mid \theta_i)}\right)$$

The prior cancels, so no integral is needed.

Auxiliary Gibbs Sampling

Introduce auxiliary draws

$$\theta_1^*, \ldots, \theta_m^* \sim G_0$$

Update assignments via

$$P(c_i = c \mid \cdot) \propto \begin{cases} n_{-i,c} F(y_i \mid \theta_c) \\ \frac{\alpha}{m} F(y_i \mid \theta_c^*) \end{cases}$$

Mixing Behavior

Standard Gibbs updates one observation at a time:

$$c_i \to c_i'$$

This leads to slow exploration of partitions and difficulty in splitting or merging clusters.

Conceptual Summary

Posterior inference targets the joint law

$$p(\Pi_n, \{\theta_c\} \mid y)$$

of the random partition and its occupied component parameters. The two main difficulties are nonconjugate component updates and the countably infinite mixture representation. Practical samplers combine collapsed Gibbs updates, Metropolis–Hastings steps for unavailable full conditionals, and auxiliary variables that expose only finitely many components at each iteration.

3.9 Sampling Techniques: Slice Sampling for DP Mixtures

Hyperpriors for Base Measure

Assume the base measure is parameterized as $\phi = (m, B)$ with priors

$$m \sim \mathcal{N}(m_0, D), \quad B \sim IGamma(a, b)$$

Given cluster parameters $\{\theta_j^*\}_{j=1}^k$, the posterior for m is

$$m \mid \cdots \sim \mathcal{N}(m_1, D_1), \quad D_1^{-1} = \frac{1}{D} + \frac{k}{B}, \quad m_1 = D_1 \left(\frac{m_0}{D} + \frac{1}{B} \sum_{j=1}^{k} \theta_j^* \right)$$

The posterior for B is

$$B \mid \cdots \sim IGamma(a_1, b_1), \quad a_1 = a + \frac{k}{2}, \quad b_1 = b + \frac{1}{2} \sum_{j=1}^{k} (\theta_j^* - m)^2$$

Sampling the Concentration Parameter

We use the marginal:

$$p(k \mid M) \propto M^k \frac{\Gamma(M)}{\Gamma(M+n)}$$

Introduce auxiliary variable:

$$\eta \mid M, k \sim Beta(M+1, n)$$

If prior $M \sim Gamma(c, d)$, then:

$$M \mid \eta, k \sim \pi \cdot Gamma(c+k, d - \log \eta) + (1-\pi) \cdot Gamma(c+k-1, d - \log \eta),$$

using the shape–rate convention, where

$$\frac{\pi}{1-\pi} = \frac{c+k-1}{n\{d-\log\eta\}}.$$

This Escobar–West augmentation replaces a nonstandard one-dimensional conditional by a beta draw followed by a two-component gamma mixture. Since M controls prior cluster growth, its posterior can be sensitive to (c, d) when n is modest; posterior summaries should include both M and the induced distribution of K_n.

No-Gaps Algorithm

Clusters are labeled consecutively:

$$c_i \in \{1, \ldots, k\}$$

with relabeling applied whenever clusters become empty.

For each observation i, remove y_i and define:

$$k^- = \#\text{ clusters without } i, \quad n_j^- = \sum_{\ell \neq i} \mathbf{1}(c_\ell = j)$$

If y_i is not a singleton:

$$P(c_i = j \mid \cdot) \propto \begin{cases} n_j^- p(y_i \mid \theta_j^*) & j \leq k^- \\ \frac{M}{k^-+1} p(y_i \mid \theta_{k^-+1}^*) & j = k^- + 1 \end{cases}$$

If y_i is a singleton:

$$P(\text{keep } c_i) = \frac{k^- - 1}{k^-}, \quad P(\text{resample}) = \frac{1}{k^-}$$

Updating Cluster Parameters

Given assignments $\{c_i\}$:

$$p(\theta_j^* \mid y) \propto G_0(\theta_j^*) \prod_{i:c_i=j} p(y_i \mid \theta_j^*)$$

If non-conjugate, sample via Metropolis-Hastings.

New cluster candidates:

$$\theta_{k+1}^* \sim G_0$$

Slice Sampling Idea

Target density:

$$\pi(x) = \frac{f(x)}{Z}, \quad Z = \int f(s)\, ds$$

Introduce auxiliary variable u:

$$u \sim Uniform(0, f(x))$$

Define joint:

$$p(x, u) = \frac{1}{Z}\mathbf{1}(0 < u < f(x))$$

Marginalizing:

$$p(x) = \int_0^{f(x)} \frac{1}{Z} du = \frac{f(x)}{Z}$$

Slice Sampling Updates

Sample:

$$u_t \sim Uniform(0, f(x_t))$$

Define slice:

$$S(u) = \{x : f(x) \geq u\}$$

Then update:

$$x_{t+1} \sim Uniform(S(u_t))$$

Slice Sampling for DP Mixtures

The likelihood is

$$p(y_i \mid \{w_h\}, \{\tilde{\theta}_h\}) = \sum_{h=1}^{\infty} w_h p(y_i \mid \tilde{\theta}_h)$$

Introduce slice variable:

$$u_i \sim Uniform(0, w_{z_i})$$

Augmented likelihood:

$$p(y_i, u_i) = \sum_{h=1}^{\infty} \mathbf{1}(u_i < w_h) p(y_i \mid \tilde{\theta}_h)$$

Key property:

$$u_i > 0 \Rightarrow \text{only finitely many } w_h > u_i$$

Indeed, if r weights exceeded u_i, then $1 = \sum_h w_h > r u_i$, so $r < 1/u_i$. With $u_{\min} = \min_i u_i$, it suffices during that sweep to instantiate components until the remaining stick mass is below $u_{\min}$. This is a random exact truncation, not a fixed approximation to the infinite prior.

Therefore,

$$\sum_{h=1}^{\infty} \to \sum_{h:w_h>u_i}$$

which is finite.

Marginalization Check

$$\int p(y_i, u_i)du_i = \sum_{h=1}^{\infty} p(y_i \mid \tilde{\theta}_h) \int_0^{w_h} du_i = \sum_{h=1}^{\infty} w_h p(y_i \mid \tilde{\theta}_h)$$

Thus the original model is recovered.

Key Insight

One sweep samples $u_i \mid z_i, w$, extends the stick only far enough to cover $u_{\min}$, updates allocations among components satisfying $w_h > u_i$, and then updates active atoms and stick variables from their full conditionals. Marginalizing the slice variables recovers the original infinite mixture, but each realized sweep performs finite computation. Very small $u_{\min}$ can still instantiate many components, so runtime is random and depends on the current allocation and weights.

This enables exact inference without truncation.

Slice Sampler: Component Indicators

Introduce latent indicators $r_i \in \{1, 2, \dots\}$ such that

$$r_i = h \quad \iff \quad y_i \text{ is generated from } \tilde{\theta}_h$$

The augmented likelihood becomes

$$p(y_i, u_i, r_i \mid \{w_h\}, \{\tilde{\theta}_h\}) = \mathbf{1}(u_i < w_{r_i})\, p(y_i \mid \tilde{\theta}_{r_i})$$

Marginalizing out r_i:

$$\sum_{h=1}^{\infty} \mathbf{1}(u_i < w_h) p(y_i \mid \tilde{\theta}_h)$$

recovers the slice-augmented likelihood.

The full joint:

$$p(y, u, r \mid \{w_h\}, \{\tilde{\theta}_h\}) = \prod_{i=1}^{n} \mathbf{1}(u_i < w_{r_i}) p(y_i \mid \tilde{\theta}_{r_i})$$

Slice Sampler: Conditional Updates

Slice variables:

$$u_i \mid r_i, \{w_h\} \sim Uniform(0, w_{r_i})$$

Component indicators:

$$P(r_i = h \mid u_i, y_i, \{\tilde{\theta}_h\}) \propto \mathbf{1}(w_h > u_i)\, p(y_i \mid \tilde{\theta}_h)$$

Only components with $w_h > u_i$ are considered.

Component parameters:

$$p(\tilde{\theta}_h \mid \{y_i : r_i = h\}) \propto G_0(\tilde{\theta}_h) \prod_{i:r_i=h} p(y_i \mid \tilde{\theta}_h)$$

Stick-breaking weights:

Let

$$w_h = v_h \prod_{k<h} (1 - v_k)$$

Then

$$v_h \mid \{r_i\} \sim Beta\left(1 + n_h,\ M + \sum_{k>h} n_k\right), \quad n_h = \sum_i \mathbf{1}(r_i = h)$$

Finite Approximation to DP

Instead of infinite mixtures:

$$G = \sum_{h=1}^{\infty} w_h \delta_{\tilde{\theta}_h}$$

approximate with finite truncation:

$$G_H = \sum_{h=1}^{H} w_h \delta_{\tilde{\theta}_h}$$

The marginal distributions are close:

$$\|p_\infty - p_H\|_1 \leq 4ne^{-(H-1)/M}$$

Thus small H can suffice when M is moderate.

Finite DP Posterior Structure

Joint posterior:

$$p(\{r_i\}, \{\tilde{\theta}_h\}, \{v_h\} \mid y) \propto \prod_{i=1}^{n} p(y_i \mid \tilde{\theta}_{r_i}) \prod_{i=1}^{n} p(r_i \mid \{v_h\}) \prod_{h=1}^{H} dG_0(\tilde{\theta}_h) \prod_{h=1}^{H-1} p(v_h \mid M)$$

The full conditional distributions are

$$p(\tilde{\theta}_h \mid \cdot) \propto G_0(\tilde{\theta}_h) \prod_{i:r_i=h} p(y_i \mid \tilde{\theta}_h)$$

$$v_h \mid \cdot \sim Beta\left(1 + n_h,\ M + \sum_{k>h} n_k\right)$$

$$P(r_i = h \mid \cdot) \propto w_h p(y_i \mid \tilde{\theta}_h)$$

Key Comparison

Slice sampler: exact, dynamically truncates via $\{u_i\}$; Finite DP: fixed truncation level H; Both reduce infinite mixture to finite computation per iteration.

3.9.1 Reading: Gibbs Sampling for Stick-Breaking Priors

Stick-Breaking Random Measures

A general stick-breaking prior can be written as

$$G = \sum_{k=1}^{\infty} p_k \delta_{Z_k}$$

where

$$p_1 = V_1, \quad p_k = V_k \prod_{l<k} (1 - V_l), \quad V_k \sim Beta(a_k, b_k)$$

The atoms satisfy

$$Z_k \overset{iid}{\sim} H$$

This defines a broad class of random probability measures including DP and Pitman–Yor.

Finite vs Infinite Construction

Finite case:

$$\sum_{k=1}^{N} p_k = 1 \quad \text{with } V_N = 1$$

Infinite case requires

$$\sum_{k=1}^{\infty} \mathbb{E}[\log(1 - V_k)] = -\infty$$

to ensure weights sum to 1 almost surely.

Dirichlet Process Special Case

DP corresponds to

$$V_k \sim Beta(1, \alpha)$$

giving

$$G \sim DP(\alpha, H)$$

This provides a constructive definition equivalent to Ferguson's original formulation.

Generalized Gibbs Sampling

Two main approaches:

Pólya urn Gibbs:

$$\theta_{n+1} \mid \theta_{1:n} \sim \frac{\alpha}{\alpha + n} H + \sum_{k} \frac{n_k}{\alpha + n} \delta_{\theta_k^*}$$

Blocked Gibbs: Directly sample

$$\{V_k\}, \{Z_k\}, \{p_k\}$$

This avoids marginalization and improves mixing.

Pitman–Yor Extension

Stick-breaking form:

$$V_k \sim Beta(1 - a, b + ka)$$

This yields richer clustering behavior than DP.

Key Insight

Stick-breaking provides:

explicit construction of random measures; direct sampling schemes; flexibility beyond Dirichlet process.

3.10 Reading: Slice Sampling Mixture Models

Dirichlet Process Mixture Model

The model:

$$f(y) = \int K(y \mid \phi)\, dP(\phi), \quad P \sim DP(M, P_0)$$

Stick-breaking form:

$$P = \sum_{j=1}^{\infty} w_j \delta_{\phi_j}, \quad w_j = z_j \prod_{l<j} (1 - z_l), \quad z_j \sim Beta(1, M)$$

Slice Sampling Idea

Introduce auxiliary variable:

$$u_i \sim Uniform(0, w_{d_i})$$

The joint distribution is

$$p(y_i, u_i) = \sum_{j=1}^{\infty} \mathbf{1}(u_i < w_j) K(y_i \mid \phi_j)$$

This restricts to finite set:

$$A_u = \{j : w_j > u_i\}$$

Latent Component Indicator

Introduce

$$d_i \in A_u$$

Then

$$p(y_i, u_i, d_i) = \mathbf{1}(u_i < w_{d_i}) K(y_i \mid \phi_{d_i})$$

This converts infinite mixture into finite conditional problem.

Gibbs Updates

$$\phi_j \mid \cdot \propto p_0(\phi_j) \prod_{i:d_i=j} K(y_i \mid \phi_j)$$

$$z_j \sim Beta\left(1 + \sum_i \mathbf{1}(d_i = j),\ M + \sum_i \mathbf{1}(d_i > j)\right)$$

$$u_i \sim Uniform(0, w_{d_i})$$

$$P(d_i = k) \propto \mathbf{1}(w_k > u_i) K(y_i \mid \phi_k)$$

Key Insight

Slice variables ensure:

only finitely many components active per iteration; exact inference without truncation; easy extension to general stick-breaking priors.

3.11 Reading: Bayesian Density Estimation via DP Mixtures

Model Setup

The observed data are

$$y_i \mid \theta_i \sim N(\mu_i, V_i)$$

The prior is

$$\theta_i \sim G, \quad G \sim DP(\alpha G_0)$$

Clustering Property

Conditional distribution:

$$\theta_i \mid \theta_{-i} \sim \frac{\alpha}{\alpha + n - 1} G_0 + \sum_{j \neq i} \frac{1}{\alpha + n - 1} \delta_{\theta_j}$$

Leads to clustering (ties among θ_i).

Posterior Predictive

$$y_{n+1} \mid y_{1:n} \sim \frac{\alpha}{\alpha + n} \int K(y \mid \theta) dG_0(\theta) + \sum_k \frac{n_k}{\alpha + n} K(y \mid \theta_k^*)$$

Interpretation

The posterior predictive distribution mixes a new component drawn through G_0 with kernels attached to occupied clusters. Its data-dependent weights provide adaptive smoothing: dense regions reinforce existing components, while the base-measure term preserves the possibility of new structure. This resembles kernel density estimation, but both the effective number of kernels and their uncertainty are inferred rather than fixed.

Gibbs Sampling Strategy

Update each θ_i:

$$p(\theta_i \mid \theta_{-i}, y) \propto \alpha p(y_i \mid \theta_i) G_0(\theta_i) + \sum_{j \neq i} p(y_i \mid \theta_j) \delta_{\theta_j}$$

This forms basis for CRP-based samplers.

Key Insight

A DP mixture simultaneously defines a flexible density estimator and a random clustering of the observations. Posterior uncertainty in the partition induces uncertainty in the represented number of components; that number should not be interpreted automatically as a consistent estimate of a true finite mixture order.

3.12 Advanced MCMC Methods

Metropolis–Hastings Algorithm

Given current state $x^{(t)}$:

$$Y_t \sim q(y \mid x^{(t)})$$

Accept with probability

$$\rho(x, y) = \min\left\{\frac{\pi(y)\, q(x \mid y)}{\pi(x)\, q(y \mid x)}, 1\right\}$$

The update is

$$x^{(t+1)} = \begin{cases} Y_t & \text{with prob } \rho(x^{(t)}, Y_t) \\ x^{(t)} & \text{otherwise} \end{cases}$$

3.12.1 Random Walk Metropolis–Hastings (RWMH)

The proposal distribution is

$$Y_t = x^{(t)} + Z_t, \quad Z_t \sim g$$

Thus

$$q(y \mid x) = g(y - x)$$

Common choices:

$$Z_t \sim N(0, \sigma^2 I_d)$$

Scaling of the Proposal

Tradeoff:

$$\sigma \text{ small} \Rightarrow \text{slow exploration}$$

$$\sigma \text{ large} \Rightarrow \text{high rejection}$$

Goal: balance via acceptance rate.

Optimal Acceptance Rates

For high dimension d:

$$\text{Optimal rate} \approx 0.234$$

For $d = 1$:

$$\text{Optimal rate} \approx 0.44$$

Problems with MH

Local Trap

Example:

$$\pi(x) = \frac{1}{3}N(\mu_1, \Sigma) + \frac{2}{3}N(\mu_2, \Sigma)$$

Chains may remain in one mode:

$$P(\text{jump between modes}) \approx \text{small}$$

Intractable Normalizing Constants

The target distribution is

$$f(x) \propto c(x)\psi(x)$$

Acceptance ratio requires:

$$\frac{c(y)}{c(x)}$$

If $c(x)$ unknown $\Rightarrow$ MH not feasible.

3.12.2 Metropolis Adjusted Langevin Algorithm (MALA)

Continuous-time:

$$dX_t = dB_t + \frac{1}{2}\nabla \log \pi(X_t)$$

Discrete proposal:

$$Y_t \sim N\left(x^{(t)} + \frac{\sigma^2}{2}\nabla \log \pi(x^{(t)}), \sigma^2 I_d\right)$$

Properties of MALA

In the standard high-dimensional product-target asymptotic regime, the optimal acceptance probability is approximately 0.574; this is a tuning guide, not a universal theorem for every target. Gradient information makes proposals follow local posterior geometry and can improve mixing, but requires a stable and computable $\nabla \log \pi(x)$.

Implementation Issues

Diagnostics should be evaluated on functionals relevant to the analysis and across dispersed chains. Efficient sampling requires exploration of all important posterior regions and low integrated autocorrelation, not merely an apparently stable trace within one mode.

Multiple Chains vs Single Chain

Multiple chains $\{x_t^{(m)}\}_{m=1}^{M}$ expose sensitivity to initialization and permit between-chain diagnostics. One long chain can estimate expectations when it mixes well, but within-chain stability alone cannot reveal an unvisited mode. Independence is unnecessary; the relevant quantity is the effective sample size after accounting for autocorrelation.

Thinning

Thinning retains $x_k, x_{2k}, \ldots$. It may reduce storage or the cost of downstream processing, but ordinarily decreases statistical efficiency because discarded draws still contain information.

Reparametrization

Reparameterization targets high posterior correlation. In the hierarchical model

$$y_{ij} = \mu + \alpha_i + e_{ij}$$

$$\alpha_i \sim N(0, \tau^2), \quad e_{ij} \sim N(0, \sigma^2)$$

the posterior correlation between μ and α_i can be close to minus one. Defining

$$\beta_i = \mu + \alpha_i$$

can reduce that dependence and improve mixing. Whether centered or noncentered coordinates are preferable depends on the relative strength of the likelihood and hierarchical prior.

Blocking

For correlated coordinates of $\theta = (\theta_1, \ldots, \theta_k)$, replace scalar conditionals by a joint block update such as

$$(\theta_{i_1}, \ldots, \theta_{i_k}) \mid \theta_{-(i_1,\ldots,i_k)}$$

so that the chain can move along correlated directions. Larger blocks may mix better but are more expensive to sample.

Collapsing

Collapsing analytically integrates nuisance variables, for example sampling (X_1, X_2) from its marginal and recovering X_3 conditionally afterward. Removing a strongly coupled variable often lowers Monte Carlo variance, though partially collapsed updates must be ordered carefully to preserve the target.

Auxiliary Variable Methods

Auxiliary-variable methods introduce u so that

$$p(x) = \int p(x, u)\, du$$

while choosing the joint law to make the full conditionals easier. Slice sampling is a principal example: a larger state space replaces evaluation of a normalizing constant or infinite sum by simple conditional restrictions.

Choosing a Computational Strategy

Efficiency depends jointly on proposal design, parameterization, and posterior geometry. Gradients, blocking, collapsing, and augmentation address different sources of dependence; none substitutes for checking multimodality and functional-specific effective sample size.

3.13 Auxiliary Variable Methods and Hybrid MCMC

Cauchy–Normal Model via Data Augmentation

$$X_i \overset{iid}{\sim} \text{Cauchy}(\mu, 1)$$

The likelihood is

$$L(\mu \mid x_{1:n}) = \prod_{i=1}^{n} \frac{1}{\pi(1 + (x_i - \mu)^2)}$$

The prior is

$$\mu \sim N(0, 10)$$

The posterior is

$$\pi(\mu \mid x) \propto e^{-\mu^2/20} \prod_{i=1}^{n} \frac{1}{1 + (x_i - \mu)^2}$$

Introduce latent variables:

$$\frac{1}{1 + (x_i - \mu)^2} = \int_0^\infty e^{-\omega_i(1+(x_i-\mu)^2)} d\omega_i$$

Augmented posterior:

$$\pi(\mu, \omega \mid x) \propto e^{-\mu^2/20} \prod_{i=1}^{n} e^{-\omega_i(1+(x_i-\mu)^2)}$$

Gibbs updates:

$$\omega_i \mid \mu \sim \text{Exp}(1 + (x_i - \mu)^2)$$

$$\mu \mid \omega \sim N\left(\frac{\sum_i \omega_i x_i}{\sum_i \omega_i + 1/20}, \frac{1}{2\sum_i \omega_i + 1/10}\right)$$

Metropolis-within-Gibbs

When a full conditional is not tractable:

$$\theta_j \mid \theta_{-j} \quad \text{sampled via MH}$$

Procedure:

$$Y \sim q(\cdot \mid \theta_j)$$

$$\text{accept with prob } \min\left\{\frac{\pi(Y \mid \theta_{-j})q(\theta_j \mid Y)}{\pi(\theta_j \mid \theta_{-j})q(Y \mid \theta_j)}, 1\right\}$$

The key idea is to

$$\text{Replace difficult Gibbs steps with MH steps}$$

3.13.1 Adaptive Gibbs Samplers

The objective is to

$$\text{improve efficiency by learning during sampling}$$

Selection probabilities:

$$\alpha_n = (\alpha_{n,1}, \ldots, \alpha_{n,d})$$

Update rule:

$$\alpha_n = R_n(\alpha_{n-1}, X_{n-1}, \ldots)$$

Transition:

$$X_n \sim P_{\alpha_n}(X_{n-1}, \cdot)$$

Key condition (diminishing adaptation):

$$|\alpha_n - \alpha_{n-1}| \to 0$$

Ensures:

$$\text{ergodicity under suitable conditions}$$

3.13.2 Random Scan Gibbs Sampler

State:

$$X = (X_1, \ldots, X_d)$$

At each step:

$$i \sim \text{Categorical}(\alpha_1, \ldots, \alpha_d)$$

$$X_i \sim \pi(\cdot \mid X_{-i})$$

The kernel is

$$P_\alpha = \sum_{i=1}^{d} \alpha_i P_i$$

Adaptive Random Scan Gibbs

The update is

$$\alpha_n = R_n(\alpha_{n-1}, X_{0:n-1})$$

Then

$$X_n \sim P_{\alpha_n}(X_{n-1}, \cdot)$$

Key result:

$$|\alpha_n - \alpha_{n-1}| \to 0 \quad \Rightarrow \quad \text{ergodicity (under conditions)}$$

Metropolis-within-Gibbs (Adaptive)

If conditional unavailable:

$$X_i \sim Q_{x_{-i}}(X_i, \cdot)$$

Accept:

$$\min\left\{1, \frac{\pi(Y \mid x_{-i}) q(Y, X_i)}{\pi(X_i \mid x_{-i}) q(X_i, Y)}\right\}$$

Adaptive version:

$$\gamma_n \text{ controls proposal}$$

$$Q_n = Q_{\gamma_n}$$

Condition:

$$\|Q_{n+1} - Q_n\| \to 0$$

Ergodicity Conditions (Key Structure)

Two critical requirements:

Diminishing adaptation:

$$\|P_n - P_{n-1}\| \to 0$$

Containment:

$$\text{chain does not escape to infinity}$$

Together imply:

$$X_n \xrightarrow{d} \pi$$

Failure of Naive Adaptation

Convergence of a tuning sequence, such as $\alpha_n \to \alpha$, does not by itself imply convergence of the adaptive chain. The transient sequence of kernels may drive the chain into regions from which it returns increasingly slowly, producing a nonergodic process even though the eventual proposal looks reasonable. Diminishing adaptation must therefore be paired with containment or another uniform stability condition.

3.13.3 Adaptive Metropolis-within-Gibbs (Full)

The method updates both a scale parameter α_n and a covariance or shape parameter γ_n, using recursions $alpha_n = R_n(\cdot)$ and $\gamma_n = R'_n(\cdot)$, before drawing $X_n \sim P_{\alpha_n,\gamma_n}(X_{n-1}, \cdot)$. The recursions must decay slowly enough to learn useful geometry but quickly enough that successive kernels become asymptotically indistinguishable.

Slice Sampling (Revisited)

The joint distribution is

$$p(x, u) = \frac{1}{Z}\mathbf{1}(0 < u < f(x))$$

Gibbs steps:

$$u \mid x \sim \text{Unif}(0, f(x))$$

$$x \mid u \sim \text{Unif}(\{x : f(x) \geq u\})$$

For DP mixtures:

$$u_i < w_{r_i} \Rightarrow \text{finite truncation}$$

Key Insight Across Methods

All of these methods seek to reduce autocorrelation through better proposals, reparameterization, or augmentation. Their common design constraint is to modify the Markov kernel while preserving the target distribution π.

Takeaways

Gibbs sampling is simple when full conditionals are tractable but can mix slowly under strong dependence. Metropolis–Hastings is more flexible but sensitive to proposal tuning; MALA uses gradients to make locally informed proposals. Adaptive methods learn useful geometry during the run but require ergodicity safeguards, while auxiliary-variable methods trade a larger state space for simpler conditional updates.

3.14 Pitman–Yor Process (PYP)

Motivation: Limitations of the Dirichlet Process

The DP induces logarithmic cluster growth, $\mathbb{E}(K_n) \sim \alpha \log n$, and its size-biased stick weights decay geometrically on average. This is appropriate for many density-estimation problems but restrictive when observed partitions contain many rare types and power-law frequency behavior, as in species sampling or language data. The Pitman–Yor process adds a discount parameter while retaining exchangeability and tractable predictive probabilities.

Idea of the Pitman–Yor Process

The discount d weakens reinforcement of every occupied cluster by the same amount and transfers the removed mass to innovation. Consequently the chance of a new cluster increases with the current number of clusters, producing polynomial rather than logarithmic cluster growth when $d > 0$.

Definition: Stick-Breaking Representation

$$G = \sum_{k=1}^{\infty} \pi_k \delta_{\phi_k}, \quad \phi_k \overset{iid}{\sim} G_0$$

$$\pi_k = v_k \prod_{\ell<k} (1 - v_\ell)$$

$$v_k \sim \text{Beta}(1 - d, \alpha + kd)$$

independently, with

$$0 \le d < 1, \quad \alpha > -d$$

in the usual infinite-species regime. Setting $d = 0$ recovers $v_k \sim \text{Beta}(1, \alpha)$ and hence the Dirichlet process.

Pitman–Yor via Discounted CRP

If K_n clusters have sizes $n_1, \ldots, n_{K_n}$, the predictive rule is

$$\mathbb{P}(z_{n+1} = k \mid z_{1:n}) = \begin{cases} \frac{n_k - d}{n+\alpha}, & k = 1, \ldots, K_n \\ \frac{\alpha + dK_n}{n+\alpha}, & \text{new cluster} \end{cases}$$

The probabilities sum to one because $\sum_k (n_k - d) + \alpha + dK_n = n + \alpha$. Multiplying these sequential factors gives the Pitman–Yor EPPF

$$p(n_1, \ldots, n_K) = \frac{\prod_{j=1}^{K-1} (\alpha + jd)}{(\alpha + 1)_{n-1}} \prod_{k=1}^{K} (1 - d)_{n_k - 1},$$

where $(a)_m$ is the rising factorial. At $d = 0$ this reduces to the CRP EPPF.

Key Properties

The discount changes reinforcement from n_k to $n_k - d$, transferring total mass dK_n from occupied clusters to the new-cluster event. Small clusters therefore retain appreciable predictive probability, while the chance of innovation rises with the number of types already observed. This mechanism produces a more diverse partition and power-law cluster frequencies; it does not literally prevent a large cluster from dominating a particular sample.

Power-Law Behavior of Weights

Independence of the stick variables gives

$$\mathbb{E}[\pi_k] = \mathbb{E}[v_k] \prod_{\ell<k} \mathbb{E}[1 - v_\ell],$$
$$\mathbb{E}[v_k] = \frac{1-d}{\alpha + kd + 1 - d}, \qquad \mathbb{E}[1 - v_\ell] = \frac{\alpha + \ell d}{\alpha + \ell d + 1 - d}.$$

Applying gamma-function asymptotics to the product yields

$$\mathbb{E}[\pi_k] \sim C_{\alpha,d}\, k^{-1/d}\,.$$

Thus the expected size-biased weights decay polynomially rather than geometrically, with larger d producing a heavier tail.

Growth of Number of Clusters

The predictive probability of a new cluster is

$$\mathbb{P}(z_{n+1} = \text{new}) = \frac{\alpha + dK_n}{n + \alpha}\,.$$

Consequently $m_n = \mathbb{E}[K_n]$ satisfies

$$m_{n+1} = m_n + \frac{\alpha + dm_n}{n + \alpha}\,.$$

For large n, the dominant balance is

$$\frac{dm_n}{dn} \approx \frac{dm_n}{n}\,,$$

and hence

$$\log m_n \approx d \log n \quad \Rightarrow \quad m_n \sim Cn^d$$

Hence the PYP has polynomial cluster growth, in contrast with the logarithmic growth $\mathbb{E}K_n \asymp \log n$ under a DP.

Interpretation

The discount parameter generalizes the DP while preserving a tractable restaurant-process predictive rule. It is useful when the observed partition has persistent rare types and power-law frequency behavior; it should not be chosen merely because the dataset is large.

Chapter 4

Extensions of the Dirichlet Process

4.1 Pitman–Yor Process: Inference and Applications

4.1.1 Inference under Pitman–Yor Process

For the hierarchical model

$$G \sim \mathrm{PYP}(\alpha, d, G_0), \quad \theta_i \mid G \sim G, \quad y_i \mid \theta_i \sim F(\theta_i)$$

integrating out G gives the collapsed assignment update

$$\mathbb{P}(z_i = k \mid z_{-i}, y) \propto \begin{cases} (n_{k,-i} - d)\, p(y_i \mid y_{k,-i}), & k \le K_{-i} \\ (\alpha + dK_{-i})\, p(y_i \mid G_0), & k = \text{new} \end{cases}$$

where the predictive likelihoods are exactly those of a DP mixture. Only the partition weights change, so conjugate PYP and DP samplers have essentially the same per-sweep cost. The discount reduces reinforcement of every occupied cluster and transfers the accumulated mass dK_{-i} to creation of new ones.

DP vs PYP: Key Differences

The DP is the boundary case $d = 0$. In general the PYP requires $0 \le d < 1$ and $\alpha > -d$, and its stick-breaking variables satisfy

$$v_k \sim \mathrm{Beta}(1 - d, \alpha + kd).$$

Whereas DP weights and cluster frequencies decay exponentially, the PYP produces power-law frequencies. Correspondingly,

$$\mathbb{E}[K_n] \sim \log n \quad \text{(DP)}, \qquad \mathbb{E}[K_n] \sim n^d \quad \text{(PYP)}$$

up to parameter-dependent constants. The predictive weights are $n_k - d$ for an occupied cluster and

$$\Pr(\text{new cluster} \mid \Pi_n) = \frac{\alpha + dK_n}{n + \alpha}.$$

When to Use Pitman–Yor

The distinction between a DP and a Pitman–Yor prior should be tied to an observable frequency spectrum. Under a DP, new clusters become increasingly rare at rate $1/n$, and the expected number of occupied clusters is logarithmic. Under a PYP with $0 < d < 1$, new-cluster probability is asymptotically of order K_n/n and K_n grows like n^d. The PYP is therefore appropriate when rare categories persist and the empirical counts display a power-law tail, as can occur for alleles, word types, or species. It is not automatically preferable for every large dataset: estimating d near zero should recover DP-like behavior, and posterior predictive checks of frequency-of-frequencies are more informative than choosing a process by application label.

4.1.2 Case Study: Genetics

For allele observations from n individuals, let n_k be the frequency of the kth observed allele. The empirical spectrum often has a few common alleles and many singletons. The model

$$G \sim \mathrm{PYP}(\alpha, d, G_0), \quad z_i \mid G \sim G$$

allows the singleton count and the number of distinct alleles to grow at a power-law rate. A direct posterior predictive diagnostic compares the observed frequency-of-frequencies $M_r = \#\{k : n_k = r\}$ with replicated M_r values; failure specifically at $r = 1, 2$ reveals whether the prior is suppressing rare alleles even when the fitted total number of types looks adequate.

4.1.3 Case Study: Topic Modeling

In a topic model, topic k is a probability distribution over the vocabulary,

$$\phi_k = (\phi_{k,1}, \phi_{k,2}, \dots), \quad \sum_v \phi_{k,v} = 1$$

and a token assigned to that topic is generated according to

$$w_i \mid z_i = k \sim \mathrm{Categorical}(\phi_k)\,.$$

Natural-language corpora exhibit a long frequency tail: a small vocabulary is extremely common, while rare word types continue to appear as the corpus grows. Assigning a Pitman–Yor prior,

$$\phi_k \sim \mathrm{PYP}(\alpha, d, G_0)\,,$$

models this behavior through power-law atom frequencies. The discount d controls the tail exponent, while α controls the overall rate at which new word types appear. Posterior predictive checks should compare both the number of distinct types and the counts of singleton and low-frequency words; matching perplexity alone does not verify the intended tail behavior.

4.2 Hierarchical Dirichlet Process (HDP)

Motivation: Multiple Related Groups

Suppose observations $\{x_{ji} : i = 1, \ldots, n_j\}$ are exchangeable within each of J groups. Each group needs its own mixture distribution, but scientific structure suggests that the same latent components can recur with different prevalences. The HDP estimates an unknown, data-adaptive component library and group-specific weights over that shared library.

Why Sharing Matters

A finite hierarchical mixture requires a fixed component count K; the HDP allows the represented count to grow with the data. Sharing atoms lets large groups estimate component-specific parameters that would be weakly identified in small groups, while group-specific weights preserve genuine prevalence differences. Thus the model pools component identities without forcing the groups to have the same distribution.

Failure of Naive Construction

One might try

$$G_j \sim \mathrm{DP}(\alpha_j, G_0(\tau)), \quad \tau \sim p(\tau)$$

with $G_j \perp G_{j'} \mid \tau$. If $G_0(\tau)$ is continuous, independent DP draws have disjoint atoms almost surely, so

$$\Pr(\theta_{jk} = \theta_{j'k'}) = 0.$$

Sharing a hyperparameter therefore correlates broad distributional features but does not share mixture components. The HDP solves this by using a discrete random base measure.

HDP Model Definition

$$G_0 \mid \gamma, H \sim \mathrm{DP}(\gamma, H)$$

$$G_j \mid \alpha_0, G_0 \sim \mathrm{DP}(\alpha_0, G_0)$$

$$\theta_{ji} \mid G_j \sim G_j, \quad x_{ji} \mid \theta_{ji} \sim F(\theta_{ji})$$

Because a DP draw G_0 is discrete, every G_j is supported on the same random atoms. The local concentration α_0 controls dispersion around G_0, while γ controls the complexity of the global component library.

4.3 Hierarchical Dirichlet Process: Constructions and Inference

4.3.1 Stick-Breaking Construction of HDP

Write the global measure as

$$G_0 = \sum_{k=1}^{\infty} \beta_k \delta_{\phi_k}, \quad \phi_k \overset{iid}{\sim} H, \quad \beta \sim \mathrm{GEM}(\gamma)$$

and draw $G_j \mid G_0 \sim \mathrm{DP}(\alpha_0, G_0)$. Since G_0 is discrete, each group has the representation

$$G_j = \sum_{k=1}^{\infty} \pi_{jk} \delta_{\phi_k}$$

with the same atoms and different weights.

Finite-Dimensional Marginals

For a measurable partition $(A_1, \ldots, A_r)$, the defining finite-dimensional law is

$$(G_j(A_1), \ldots, G_j(A_r)) \sim \mathrm{Dir}(\alpha_0 G_0(A_1), \ldots, \alpha_0 G_0(A_r)) .$$

If $K_\ell = \{k : \phi_k \in A_\ell\}$, then

$$G_j(A_\ell) = \sum_{k \in K_\ell} \pi_{jk}, \quad G_0(A_\ell) = \sum_{k \in K_\ell} \beta_k ,$$

so Dirichlet aggregation gives

$$\left(\sum_{k \in K_1} \pi_{jk}, \ldots, \sum_{k \in K_r} \pi_{jk} \right) \sim \mathrm{Dir} \left(\alpha_0 \sum_{k \in K_1} \beta_k, \ldots \right) .$$

This calculation makes precise the statement that every group uses the same atoms but may assign them different total mass.

Stick-Breaking for Group Weights

The group weights may be exposed sequentially as

$$\pi_{jk} = \pi'_{jk} \prod_{\ell < k} (1 - \pi'_{j\ell}) \quad \text{with} \quad \pi'_{jk} \sim \mathrm{Beta} \left(\alpha_0 \beta_k, \alpha_0 \sum_{\ell > k} \beta_\ell \right) .$$

This is not an independent GEM construction: the beta parameters depend on the global weight vector β. It follows from aggregating the Dirichlet law into the first $k-1$ atoms, atom k, and the remaining tail, followed by renormalizing the last two masses. Consequently $\mathbb{E}[\pi_{jk} \mid \beta] = \beta_k$, as required by $\mathbb{E}[G_j \mid G_0] = G_0$.

4.3.2 Chinese Restaurant Franchise (CRF)

The Chinese restaurant franchise is the partition-valued representation of

$$G_0 \sim \mathrm{DP}(\gamma, H), \quad G_j \sim \mathrm{DP}(\alpha_0, G_0)\,.$$

Group j is represented by a restaurant, its observations by customers, and its local clusters by tables. Each table orders one global dish. Tables in different restaurants can order the same dish, which is exactly how the HDP shares a component parameter without forcing the local partitions to agree.

CRF Variables

Let $\phi_k \sim H$ denote the parameter of global dish k. If table t in restaurant j orders dish k_{jt}, then $\psi_{jt} = \phi_{k_{jt}}$; if customer i sits at table t_{ji}, then

$$\theta_{ji} = \psi_{j,t_{ji}}, \qquad x_{ji} \mid \theta_{ji} \sim F(\theta_{ji}).$$

The two indices are necessary: t_{ji} describes within-group clustering, whereas k_{jt} links local clusters across groups.

Predictive Distribution for Customers

Integrating out G_j:

$$\theta_{ji} \mid \theta_{j,1:i-1} \sim \sum_{t=1}^{m_j} \frac{n_{jt}}{i-1+\alpha_0}\delta_{\psi_{jt}} + \frac{\alpha_0}{i-1+\alpha_0}G_0$$

Thus a customer joins table t with probability proportional to its current occupancy n_{jt} and starts a new table with probability proportional to α_0. The parameter α_0 therefore controls local, not global, partition complexity.

Predictive Distribution for Tables

Integrating out G_0:

$$\psi_{jt} \mid \{\psi\}_{-jt} \sim \sum_{k=1}^{K} \frac{m_k}{m_\cdot + \gamma}\delta_{\phi_k} + \frac{\gamma}{m_\cdot + \gamma}H$$

Here m_k is the number of tables, across all restaurants, serving dish k, and $m_\cdot = \sum_k m_k$. A new table reuses dish k with probability proportional to m_k and creates a new dish with probability proportional to γ. Notice that global popularity is counted in tables rather than customers; a large group cannot dominate the global process merely by seating many customers at one table.

Index-Based Sampling View

Customer → table:

$$\mathbb{P}(t_{ji} = t \mid \cdot) = \begin{cases} \frac{n_{jt}}{i-1+\alpha_0}, & t \le m_j \\ \frac{\alpha_0}{i-1+\alpha_0}, & t = m_j + 1 \end{cases}$$

Table → dish:

$$\mathbb{P}(k_{jt} = k \mid \cdot) = \begin{cases} \frac{m_k}{m_\cdot+\gamma}, & k \le K \\ \frac{\gamma}{m_\cdot+\gamma}, & k = K + 1 \end{cases}$$

Collapsed Gibbs Sampling: Table Assignment Update

For customer i in group j, the table assignment update has the Gibbs form

$$p(t_{ji} = t \mid \text{rest}, x) \propto p(t_{ji} = t \mid \text{rest})\, p(x_{ji} \mid t_{ji} = t, \text{rest})$$

The prior term comes from the CRP within group j:

$$p(t_{ji} = t \mid t^{-ji}) \propto \begin{cases} n_{jt}^{-ji}, & \text{if table } t \text{ exists,} \\ \alpha_0, & \text{if table } t \text{ is new.} \end{cases}$$

The likelihood term is collapsed by integrating out the dish parameter ϕ_k.

Collapsed Predictive Density

Define the collapsed predictive density

$$f_k^{-x_{ji}}(x_{ji}) = p(x_{ji} \mid \{x_{j'i'} : z_{j'i'} = k,\ (j', i') \neq (j, i)\}).$$

Using Bayes' rule,

$$f_k^{-x_{ji}}(x_{ji}) = \frac{\int f(x_{ji} \mid \phi_k) \prod_{j'i' \neq ji,\, z_{j'i'}=k} f(x_{j'i'} \mid \phi_k) h(\phi_k)\, d\phi_k}{\int \prod_{j'i' \neq ji,\, z_{j'i'}=k} f(x_{j'i'} \mid \phi_k) h(\phi_k)\, d\phi_k}.$$

h is the density of the base measure H. The denominator is the marginal likelihood of observations currently assigned to global cluster k, excluding x_{ji}. The numerator is the marginal likelihood if x_{ji} were added to cluster k. If $h(\phi)$ is conjugate to $f(x \mid \phi)$, this predictive density is available in closed form.

Collapsed Gibbs Sampling: Dish Assignment Update

For a newly created table t in group j, the dish assignment update is

$$p(k_{jt} = k \mid \text{rest}, x) \propto p(k_{jt} = k \mid \text{rest})\, p(x_{jt} \mid k_{jt} = k, \text{rest}).$$

The prior term comes from the global CRP:

$$p(k_{jt} = k \mid k^{-jt}) \propto \begin{cases} m_{\cdot k}, & \text{if dish } k \text{ exists,} \\ \gamma, & \text{if dish } k \text{ is new.} \end{cases}$$

The likelihood term is collapsed over ϕ_k. If table t is assigned dish k, all customers at that table share parameter ϕ_k:

$$p(x_{jt} \mid k, \text{rest}) = \prod_{i:t_{ji}=t} f_k^{-x_{ji}}(x_{ji}).$$

Since the table was just created, it contains the single customer that triggered the new table:

$$\prod_{i:t_{ji}=t} f_k^{-x_{ji}}(x_{ji}) = f_k^{-x_{ji}}(x_{ji}).$$

Combining the prior and predictive factors yields

$$p(k_{jt} = k \mid \cdot) \propto \begin{cases} m_{\cdot k}\, f_k^{-x_{ji}}(x_{ji}), & \text{existing dish } k, \\ \gamma\, f_{\text{new}}^{-x_{ji}}(x_{ji}), & \text{new dish.} \end{cases}$$

4.3.3 Case Study: BNP Modeling of Marathon Data

The marathon application treats each age–gender stratum as a group and each runner's normalized split-time vector as a compositional observation,

$$x_{ji} = (\text{fraction of time spent in each segment}).$$

An HDP shares a global library of pace profiles while allowing their prevalence to vary across groups:

$$G_0 \sim \text{DP}(\gamma, H), \qquad G_j \mid G_0 \sim \text{DP}(\alpha, G_0), \qquad G_j = \sum_k \pi_{jk} \delta_{\phi_k},$$

$$x_{ji} \mid z_{ji} = k \sim \text{Dirichlet}(\tau \phi_k)$$

with ϕ_k a simplex-valued global pace pattern, τ controlling within-pattern dispersion, and π_{jk} the prevalence in group j. For the NYC Marathon data, posterior inference can distinguish steady pacing from a fast-start/slow-finish profile while borrowing information across strata. Given partial splits for a new runner, posterior prediction averages the remaining-time distribution over both the uncertain pattern assignment and pattern parameters. This is a concrete advantage over assigning a runner to a single estimated cluster before prediction.

4.4 Dependent Dirichlet Processes (DDP)

Motivation

A single Dirichlet process describes one random distribution. Regression and longitudinal applications instead require a collection $\mathcal{G} = \{G_x : x \in \mathcal{X}\}$ whose members vary with a predictor, time, or group index. A dependent Dirichlet process is a joint prior on this collection that preserves DP-like marginal flexibility while inducing strong dependence between G_x and $G_{x'}$ when x and x' are similar.

DDP Construction

One common-weights construction is

$$G_x = \sum_h p_h \delta_{m_h(x)}$$

with $(p_h) \sim \text{GEM}(\alpha)$ and independent stochastic processes $m_h(\cdot)$. Component h then retains its identity across predictor values while its location moves. A common-atoms construction instead uses $G_x = \sum_h p_h(x)\delta_{\theta_h}$, so prevalence rather than location changes with x.

Types of Dependence

Dependence may therefore enter through atoms, weights, or both. These choices are scientifically different: atom dependence describes a population whose latent subgroups change character with x, whereas weight dependence describes stable subgroup types whose frequencies change with x.

Example: Health Outcomes

Let Y be a continuous health outcome, x_1 an exposure, and $x_{2:p}$ confounders. Mean regression can miss an exposure effect that changes only a lower tail or creates a second mode. A dependent random measure models the full conditional law,

$$Y \mid x \sim G_x,$$

so posterior summaries may include $\Pr(Y < c \mid x)$, conditional quantiles, and tail-risk contrasts without dichotomizing the observed outcome before analysis.

Example 4.4.1. Suppose exposed and unexposed groups have the same conditional mean gestational age, but exposure moves ten percent of its mass from 39 weeks to 34 weeks. A mean model can show little change, whereas a DDP mixture can represent the additional preterm mode. The clinically relevant contrast

$$\Pr(Y < 37 \mid x_1 = 1, x_{2:p}) - \Pr(Y < 37 \mid x_1 = 0, x_{2:p})$$

is computed for every posterior draw of the two conditional distributions.

DDE and Preterm Birth Example

A conventional analysis dichotomizes gestational age and fits

$$\operatorname{logit} \Pr(Y_i < 37 \mid x_i) = \beta_0 + \sum_{h=1}^{4} \beta_h \, dde_{hi} + z_i^\top \alpha$$

for exposure indicators dde_{hi} and confounders z_i. This estimates a preterm-birth odds ratio but discards ordering and spacing. A DDP mixture retains the continuous response and recovers the same risk as the posterior functional $G_x((-\infty, 37))$, while also estimating quantiles, modes, and the entire dose-dependent tail.

Key Insight of DDP

The inferential object is $G_x = \mathcal{L}(Y \mid X = x)$, not merely $\mathbb{E}(Y \mid X = x)$. Changes in variance, modality, skewness, and tail probability are consequently estimable without separate, mutually incompatible models for each feature.

Mixture Models and Regression Extensions

Basic Mixture Model

Let y_i denote gestational age for individual i. Model density as a mixture:

$$f(y_i) = \int \mathcal{N}(y_i; \mu, \sigma^2)\, dG(\mu, \sigma^2)$$

Finite mixture approximation:

$$G = \sum_{h=1}^{k} p_h \delta_{\theta_h}, \quad \theta_h = (\mu_h, \sigma_h^2)$$

Mixtures of Gaussians can approximate flexible densities.

Mixtures of Regression Models

Incorporate predictors x_i (e.g., DDE):

$$f(y_i \mid x_i) = \sum_{h=1}^{k} p_h \mathcal{N}(y_i; x_i'\beta_h, \sigma_h^2)$$

Hierarchical mixture of experts (HME): Replace weights p_h with $p_h(x_i)$; Allows predictor-dependent clustering. Typically estimated via EM: Can overfit; Requires choosing k.

Infinite Mixtures Motivation

Avoid fixing number of components k; Introduce nonparametric prior:

$$G_x = \{G_x : x \in \mathcal{X}\}$$

If G is global, use:

$$G \sim \mathrm{DP}(\alpha G_0)$$

4.4.1 Dependent Dirichlet Processes (DDP)

General Form

Define predictor-dependent random measure:

$$G_x = \sum_{h=1}^{\infty} p_h \delta_{\theta_h(x)}$$

Atoms vary with x:

$$\theta_h(\cdot) \sim \text{Gaussian Process}$$

Weights may be: shared across x; or predictor-dependent.

Key Idea

Introduce dependence across x through: weights $p_h(x)$, or; locations $\theta_h(x)$. Goal: flexible conditional distributions $f(y \mid x)$.

4.4.2 Weighted Mixture of Dirichlet Processes

Construct local DPs at knots x_j^*:

$$G_j^* \sim \mathrm{DP}(M, G_0)$$

Define

$$G_x(B) = \sum_{j=1}^{J} \frac{K(x, x_j^*)}{\sum_\ell K(x, x_\ell^*)} G_j^*(B)$$

This means that Nearby knots receive higher weight; Smooth variation across predictor space.

4.4.3 Kernel Stick-Breaking Process (KSBP)

Construction

Define

$$G_x = \sum_{h=1}^{\infty} \left[V_h K(x, \Gamma_h) \prod_{\ell<h} (1 - V_\ell K(x, \Gamma_\ell)) \right] G_h^*$$

The model components are $V_h \sim \mathrm{Beta}(a_h, b_h)$; $\Gamma_h \sim H$ (locations) $G_h^* \sim \mathrm{DP}(\alpha G_0)$. $K(x, \Gamma_h)$ is a bounded kernel.

Interpretation

Weights depend on distance between x and Γ_h; Nearby components get larger weights; Induces smooth dependence across predictors.

Special Cases

If $K(x, \Gamma_h) = 1$: recovers standard DP or Pitman–Yor (depending on V_h) If $K(x, \Gamma_h) = \exp(-\psi \|x - \Gamma_h\|)$: weights decay with distance.

4.4.4 Predictor-Dependent Clustering

Marginal predictive:

$$(\phi_i \mid \phi^{(i)}, X) = \frac{\alpha}{\alpha + w_i} G_0 + \sum_{j \neq i} \frac{w_{ij}}{\alpha + w_i} \delta_{\phi_j}$$

Weights w_{ij} depend on similarity in predictors; Key difference from DP: clustering depends on x.

Hierarchical Dirichlet Process (HDP) Revisited

The model is

$$G_0 \sim \mathrm{DP}(\gamma, H), \quad G_j \mid G_0 \sim \mathrm{DP}(\alpha, G_0)$$

This means that G_0 defines global atoms; G_j selects subsets with group-specific weights. Enables sharing across groups without explicit regression.

4.4.5 Example: Information Retrieval

Words:

$$x_i \sim G_j$$

G_j: document-specific distribution; G_0: global vocabulary distribution; Topics shared across documents.

4.4.6 Nested Dirichlet Process

The model is G_j themselves are clustered; $G_j \sim Q$, where Q is a DP over distributions. Two levels: clustering of groups (e.g., states) clustering within groups (e.g., hospitals) Allows discovery of higher-level structure.

4.4.7 ANOVA Dependent Dirichlet Process (ANOVA-DDP)

Model Construction

Consider categorical predictors:

$$x_j = (v, w)$$

(e.g., row/column effects in ANOVA) Introduce dependence across x via ANOVA structure:

$$m_{xh} = \mu_h + \alpha_{h,v} + \beta_{h,w}$$

Each mixture component varies with factor levels while sharing global structure.

Equivalent Representation

The random measure is

$$G_x(\theta) = \sum_h w_h \delta_{m_{xh}}$$

Can be written as:

$$\theta_{xi} = \Phi_i' d_x$$

where d_x = design vector (ANOVA encoding) $M_h = (\mu_h, \alpha_{1,h}, \ldots, \alpha_{V,h}, \beta_{1,h}, \ldots, \beta_{W,h})$. This means that ANOVA-DDP = DP mixture of ANOVA models.

4.4.8 Problem Setting for BNP Regression

The observed data are

$$\{(y_i, x_i)\}_{i=1}^n, \quad x_i \in \mathbb{R}^Q$$

The model is

$$y_i = f(x_i) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

The objective is to Learn a distribution over functions $p(f)$.

Applications

Regression (curve fitting, smoothing) Classification (categorical outputs) Prediction (generalization)

4.4.9 Gaussian Process (GP)

Definition

A Gaussian process is a distribution over functions:

$$f \sim \mathrm{GP}(m(x), K(x, x'))$$

Any finite collection $\{f(x_1), \ldots, f(x_n)\}$ is jointly Gaussian.

Covariance Structure

Covariance matrix:

$$K_{ij} = K(x_i, x_j)$$

This means that

$$K(x_i, x_j) = \mathbb{E}[f(x_i)f(x_j)]$$

Must be positive semidefinite.

4.4.10 Common Covariance Functions

Squared Exponential (SE)

The squared-exponential kernel is

$$K(x, x') = \sigma_0^2 \exp\left(-\frac{1}{2}\left(\frac{x - x'}{\lambda}\right)^2\right).$$

It is stationary and produces almost surely infinitely differentiable sample paths. The amplitude σ_0 controls marginal variation, while the lengthscale λ controls the distance over which function values remain strongly correlated.

Matérn Class

The Matérn family is

$$K(x,x') = \frac{2^{1-\nu}}{\Gamma(\nu)}\left(\frac{\sqrt{2\nu}|x-x'|}{\lambda}\right)^{\nu} K_\nu\left(\frac{\sqrt{2\nu}|x-x'|}{\lambda}\right)$$

where K_ν on the right is the modified Bessel function. The parameter ν controls path regularity: $\nu = 1/2$ gives the exponential kernel, while $\nu \to \infty$ approaches the squared-exponential kernel.

Other Kernels

A linear trend can be represented by $K(x,x') = \sigma_0^2 + xx'$, whereas $K(x,x') = \min(x,x')$ is the nonstationary covariance of Brownian motion. A common stationary periodic kernel is

$$K(x,x') = \exp\left(-\frac{2\sin^2((x-x')/2)}{\lambda^2}\right).$$

The appropriate choice encodes prior structural knowledge: linearity, rough accumulation, or recurrence.

4.4.11 Combining Kernels

Positive-semidefinite kernels are closed under nonnegative sums and products, so $K_1 + K_2$ represents additive latent effects and $K_1 K_2$ represents an interaction. More generally, linear smoothing preserves positive semidefiniteness: if h is an integral operator, then

$$K(x,x') = \int h(x,z)K(z,z')h(x',z')\,dz\,dz'.$$

These closure rules permit complex covariance models to be assembled while retaining a valid GP prior.

4.4.12 GP Regression

Let $f \sim \mathrm{GP}(m,K)$ and observe noisy function values according to

$$y \mid f \sim \mathcal{N}(f(X), \sigma^2 I)$$

at inputs $X = (x_1,\ldots,x_n)$. Joint Gaussian conditioning implies

$$f \mid y \sim \mathrm{GP}(m_p, K_p)$$

with posterior mean

$$m_p(x) = m(x) + K(X,x)^T(K+\sigma^2 I)^{-1}(y - m(X))$$

and covariance

$$K_p(x,x') = K(x,x') - K(X,x)^T(K+\sigma^2 I)^{-1}K(X,x')$$

where K in the inverse denotes $K(X,X)$. The mean is a kernel-weighted linear smoother. The covariance reduction depends on the input locations and noise variance, but not on the observed response values.

4.4.13 Prediction (Multi-step)

More generally, suppose observed and future responses are jointly Gaussian:

$$\begin{bmatrix} y \\ y^* \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \right)$$

Gaussian conditioning gives

$$\mathbb{E}[y^* \mid y] = \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(y - \mu_1)$$

and

$$\mathrm{Var}(y^* \mid y) = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$$

For multistep prediction, y^* is a vector and the block formula should be used jointly; forecasting one time at a time would discard posterior dependence among future values.

4.4.14 Learning Hyperparameters

Marginal likelihood:

$$p(y \mid X, \theta) = \mathcal{N}(0, K_\theta + \sigma^2 I)$$

Log marginal likelihood:

$$\log p(y \mid X, \theta) = -\frac{1}{2}\log |K_\theta + \sigma^2 I| - \frac{1}{2} y^T (K_\theta + \sigma^2 I)^{-1} y + \text{const}$$

Estimation: Maximum likelihood; Fully Bayesian (e.g., HMC)

4.4.15 Remarks

GPs average over functions, learn kernel structure from data, and provide predictive uncertainty without an explicit finite feature map. Compared with a standard SVM, the construction is probabilistic and regularization is encoded by the covariance prior, although exact GP inference is substantially more expensive.

4.4.16 Feature Selection

Represent a subset by $m \in \{0,1\}^D$. Selecting

$$\hat{m} = \arg\max_m p(D \mid m)$$

requires searching 2^D models and ignores selection uncertainty. Bayesian model averaging instead assigns posterior mass across subsets and averages their predictions; a single selected model is needed only when the downstream decision explicitly requires a sparse set.

4.4.17 Feature Selection in Bayesian Models

Feature selection is not automatically required to prevent overfitting in a fully Bayesian analysis: posterior averaging already shrinks and averages over uncertain effects. The relevant question is instead whether excluding a variable improves the decision problem. Selection is compelling when measurements are costly, interpretability is itself an objective, or prediction with all variables is computationally prohibitive. If none of these costs is present, a well-regularized Bayesian model can predict well while retaining all features and propagating uncertainty about their relevance.

Automatic Relevance Determination (ARD)

ARD uses a kernel with dimension-specific length scales,

$$K(x_n, x_{n'}) = \nu \exp\left(-\frac{1}{2}\sum_{d=1}^{D}\left(\frac{x_n^{(d)} - x_{n'}^{(d)}}{r_d}\right)^2\right)$$

where r_d controls variation along coordinate d. As $r_d \to \infty$, the kernel becomes insensitive to that coordinate, providing continuous relevance shrinkage rather than a hard inclusion decision. Because length scales can be weakly identified in finite samples, posterior uncertainty in r_d should be reported rather than converting a single estimate into a definitive selected set.

4.4.18 Limitations of Gaussian Processes

Exact factorization of the $n \times n$ covariance matrix costs $O(n^3)$ time and $O(n^2)$ memory. Model quality also depends on the covariance family: an RBF prior places mass on extremely smooth functions and can blur discontinuities or abrupt regime changes. This is a limitation of that kernel choice rather than a universal inability of Gaussian processes.

4.4.19 Scalable Gaussian Processes

Global Approximations

Use subset of size $m \ll n$: random subsampling; clustering (e.g., centroids) active learning selection. Low-rank approximation:

$$K_{nn} \approx K_{nm} K_{mm}^{-1} K_{mn}$$

Sparse Kernels

Compact support kernels:

$$K(x_i, x_j) = 0 \quad \text{if } \|x_i - x_j\| > c$$

Challenges: maintain positive semidefiniteness. Complexity:

$$\mathcal{O}(\alpha n^3)$$

with sparsity factor α.

Sparse GP (Inducing Points)

Introduce $M \ll N$ inducing points; Approximate prior:

$$p(f) \approx \mathcal{N}(0, K_{NM}K_{MM}^{-1}K_{MN} + \Lambda)$$

Benefits: Faster inversion:

$$\mathcal{O}(M^2N) \ll \mathcal{O}(N^3)$$

Efficient prediction:

$$\mathcal{O}(M) \text{ mean}, \quad \mathcal{O}(M^2) \text{ variance}$$

4.4.20 CART: Regression Trees

Motivation: Simple global models fail on complex data. Idea: Partition input space; Fit simple models locally. Implementation: Recursive binary splits; Optimize fit subject to complexity constraints.

4.4.21 Bayesian Regression Trees

The model is

$$y(x) = g(x; T, M) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

The component parameters follow T: tree structure; M: leaf parameters. Prior factorization:

$$\pi(T, M, \sigma^2) = \pi(M \mid T, \sigma^2)\pi(T \mid \sigma^2)\pi(\sigma^2)$$

4.4.22 Bayesian Additive Regression Trees (BART)

The model is

$$y(x) = \sum_{j=1}^{m} g(x; T_j, M_j) + \epsilon$$

This means that Ensemble of weak learners; Each tree captures partial structure. The prior is

$$\pi((T_1, M_1), \ldots, (T_m, M_m), \sigma^2) = \prod_{j=1}^{m} \pi(M_j \mid T_j, \sigma^2)\pi(T_j \mid \sigma^2)\pi(\sigma^2)$$

4.4.23 Regression Tree Representation

Tree defines mapping:

$$g(x; T, M) \mapsto \mu(x)$$

Leaf parameters:

$$M = (\mu_1, \ldots, \mu_b)$$

Possible leaf models: Constant; Linear; Gaussian process.

4.4.24 MCMC for Tree Models

Sampling scheme: Sample tree structure T via MH; Sample leaf parameters M via Gibbs; Sample variance σ^2 via Gibbs.

Tree Proposals (MH)

Grow: split a terminal node; Prune: collapse a subtree; Change: modify split rule; Swap: exchange split rules.

Ensures reversibility of the Markov chain.

4.4.25 Pros and Cons

Advantages

Highly flexible function approximation; Handles mixed data types naturally; Scales via parallel MCMC.

Limitations

Poor mixing (due to discrete tree moves) Limited interpretability; Can underrepresent uncertainty.

4.5 Random Partition Models

Random Probability Measures and Clustering

In Bayesian nonparametrics, we place priors on random distributions:

$$F \sim p(F), \quad F = \sum_{h=1}^{\infty} w_h \delta_{m_h}$$

Many such priors (e.g., DP) are discrete $\Rightarrow$ induce ties. Implied clustering: Draw $\theta_i \sim F$, i.i.d. Positive probability that $\theta_i = \theta_{i'}$; This defines clusters via shared values. Write the partition as

$$S_j = \{i : \theta_i = \theta_j^*\}, \quad j = 1, \ldots, k$$

Applications: Text: words grouped into topics; Biostatistics: patients grouped into subpopulations.

4.5.1 Inference on Partitions

If F is not of direct interest:

$$\rho = \{S_1, \ldots, S_k\}$$

Prefer specifying a prior directly on partitions:

$$p(\rho)$$

Notation for Random Partitions

Units: $i \in S = \{1, \ldots, n\}$; Write the partition as

$$\rho_n = \{S_1, \ldots, S_k\}, \quad S = \bigcup_{j=1}^{k} S_j$$

Cluster sizes:

$$n_j = |S_j|$$

Allocation representation:

$$s_i = j \iff i \in S_j$$

Covariate-dependent partitions:

$$p(\rho_n \mid x)$$

4.5.2 Sampling Model

Conditional on partition:

$$p(y^n \mid \rho, \theta) = \prod_{j=1}^{k} \prod_{i \in S_j} p(y_i \mid \theta_j)$$

The model components are Cluster parameters θ_j; Prior on θ_j: often conjugate; Prior on ρ: defines clustering structure.

4.5.3 Product Partition Model (PPM)

The partition prior is

$$p(\rho_n) \propto \prod_{j=1}^{k} c(S_j)$$

$c(S_j)$: cohesion function measuring cluster strength; Examples: Dirichlet process:

$$c(S_j) = \alpha(n_j - 1)!$$

Contiguity (changepoint models):

$$c(S_j) = p^{n_j - 1}(1 - p)\mathbf{1}\{S_j \text{ contiguous}\}$$

Conjugacy: Posterior remains PPM with updated cohesion.

4.5.4 Species Sampling Models (SSM)

Partition probability depends only on cluster sizes:

$$p(\rho_n) = p(|S_1|, \ldots, |S_k|)$$

Exchangeable partition probability function (EPPF) Predictive characterization (PPF):

$$p(s_{i+1} = j \mid s_1, \ldots, s_i)$$

Must satisfy consistency conditions for exchangeability.

Polya Urn (Dirichlet Process)

Let k_i be number of clusters among first i observations; Predictive rule:

$$p(s_{i+1} = \begin{cases} j & \text{existing cluster} \\ k_i + 1 & \text{new cluster} \end{cases}) = \begin{cases} \frac{n_j}{M+i} \\ \frac{M}{M+i} \end{cases}$$

Equivalent to: DP prior on G; Special case of PPM and SSM.

Model-Based Clustering

Mixture model:

$$p(y_i \mid k, \theta, \pi) = \sum_{j=1}^{k} \pi_j f_j(y_i \mid \theta_j)$$

Latent assignment:

$$p(y_i \mid s_i = j, \theta) = f_j(y_i \mid \theta_j), \quad \Pr(s_i = j) = \pi_j$$

Induces partition via latent variables s_i.

4.5.5 Covariate-Dependent Clustering

Motivation

In covariate-dependent clustering, the partition is allowed to depend on $x_1, \ldots, x_n$. Subjects with similar covariates receive greater prior probability of co-clustering, while the response likelihood determines whether that prior grouping is supported by the observed outcomes.

4.5.5.1 Example 1: Sarcoma Trial

Consider a sarcoma trial involving twelve cancer subtypes, with prognosis category available as a subtype-level covariate and subtype-specific treatment success probabilities as the estimands. Separate analyses are unstable because each subtype contains little data, whereas complete pooling erases genuine heterogeneity. A single regression surface can also be too restrictive. A covariate-dependent partition offers adaptive partial pooling: subtypes with similar prognoses are encouraged, but not forced, to share a success probability, and the posterior determines the effective clustering.

4.5.5.2 Example 2: Patient Clustering

Suppose patients are to be clustered by longitudinal response trajectories. Baseline covariates should influence which trajectories are considered similar, but the fitted mechanism must also assign a new patient to a predictive distribution. A purely response-based partition ignores useful baseline information, whereas clustering the concatenated response and covariate vector can let the arbitrary scale of the covariates dominate the partition.

Augmented Response Approach

The direct augmentation

$$\tilde{y}_i = (y_i, x_i)$$

requires a joint likelihood for scientifically different variables and may model the covariate distribution merely to induce clustering. A more targeted construction specifies the covariate-dependent partition law $p(\rho \mid x)$ directly, separating the mechanism that forms clusters from the response model used within them.

Mixtures with Covariate-Dependent Weights

A mixture-of-experts formulation is

$$y_i \mid x_i \sim \sum_j \pi_j(x_i) f_j(y_i \mid \theta_j)$$

where a latent allocation s_i has probabilities $\pi_j(x_i)$. Covariates therefore affect cluster prevalence while component parameters describe the response distribution. A conventional formulation fixes the number of components; BNP versions replace that finite gate by an infinite, predictor-indexed random measure.

Dependent Dirichlet Process (DDP)

One such representation is

$$G_x = \sum_h \pi_h(x) \delta_{\theta_h}$$

with shared atoms θ_h and predictor-dependent weights $\pi_h(x)$. Two nearby patients may then have similar allocation probabilities without being forced into the same cluster, and prediction at a new x follows directly from the corresponding weights.

4.5.6 DDP-like Constructions

Probit Stick-Breaking Process

The weights satisfy

$$\pi_h(x) = \Phi(\eta_h(x)) \prod_{\ell<h} \{1 - \Phi(\eta_\ell(x))\}$$

$\eta_h(x)$ controls dependence on covariates.

Weighted Mixture of DPs

Combine local DPs:

$$G_x(B) = \sum_{l=1}^{L} \frac{\gamma_l f(x, x_l^*)}{\sum_{l'} \gamma_{l'} f(x, x_{l'}^*)} G_l^*(B)$$

Shared atoms, location-dependent weights.

Product Partition with Covariates

Modify cohesion:

$$p(\rho) \propto \prod_{j=1}^{k} c(S_j) \exp\{t \cdot g(S_j)\}$$

$g(S_j)$ measures similarity within cluster; Example:

$$g(S_j) = \frac{2}{|S_j|(|S_j|-1)} \sum_{i<i'} \left(1 - \frac{d_{ii'}}{(1+\epsilon)d^*}\right)$$

Encourages clustering of nearby points.

Summary

Random partitions provide a direct way to model clustering; Key frameworks: PPM: cohesion-based; SSM: size-based; DP: Polya urn; Model-based: latent mixtures. Extensions: Covariate-dependent partitions; Dependent random measures; Attraction-based clustering.

4.5.7 Item Allocation and Attraction-Based Clustering

Modify predictive probability function (PPF):

$$p(s_{i+1} \mid s_1, \ldots, s_i) = \begin{cases} j & \propto \frac{1}{M+i}\, q(i+1, S_j) \\ k_i + 1 & \propto \frac{M}{M+i} \end{cases}$$

Attraction function:

$$q(i, S) = \sum_{\ell \in S} a_{i\ell}$$

Define similarity weights: Distance d_{ij} between items; $a_{ij} \propto [(1+\epsilon)d^* - d_{ij}]^T$; Normalize: $\sum_{j \neq i} a_{ij} = n - 1$. It follows that Encourages clustering of nearby points; Induces ergodic Markov chain (irreducible, aperiodic, recurrent)

4.5.8 PPM with Covariates (PPMx)

The objective is to

$$p(\rho) \to p(\rho \mid x)$$

The partition prior is

$$p(\rho_n \mid x^n) \propto \prod_{j=1}^{k} g(x_j^*)\, c(S_j)$$

$g(x_j^*)$: similarity function for cluster S_j; One construction is

$$g(x_j^*) = \int \prod_{i \in S_j} q(x_i \mid \xi_j) q(\xi_j)\, d\xi_j$$

This means that Measures within-cluster covariate homogeneity; Penalizes heterogeneous clusters.

Properties of PPMx

Symmetry:

$$p(\rho_n \mid x^*) \text{ invariant to permutations}$$

Coherence across n:

$$p(\rho_n \mid x^n) = \sum_{s_{n+1}} \int p(\rho_{n+1} \mid x^{n+1}) q(x_{n+1} \mid x^n)\, dx_{n+1}$$

Requires:

$$q(x_{n+1} \mid x^n) = \frac{g_{n+1}(x^{n+1})}{g_n(x^n)}$$

Choice of Similarity Models

Continuous covariates:

$$q(x_i \mid \xi_j) = \mathcal{N}(\xi_j, V)$$

Categorical: Multinomial likelihood; Dirichlet prior. Ordinal: Multinomial probit models. Count data:

$$q(x_i \mid \xi_j) = \text{Poisson}(\xi_j), \quad q(\xi_j) = \text{Gamma}(a, b)$$

Posterior Inference under PPMx

Augmented model:

$$q(y^n, x^n, \theta, \xi, \rho_n) = \prod_j \prod_{i \in S_j} p(y_i \mid \theta_j, x_i) q(x_i \mid \xi_j)\, p(\theta_j)\, q(\xi_j)\, c(S_j)$$

The posterior is

$$p(\rho_n \mid x^n, y^n) \propto p(y^n, x^n \mid \rho_n)$$

Posterior computation proceeds by Reduces to standard clustering after augmentation; Straightforward Gibbs updates. A crucial warning is that Joint modeling of (x, y) is not equivalent to PPMx.

4.5.9 Example: Mixed Effects Model

The observed data are $n = 47$ patients; repeated measures y_{ij}; covariate x_i (dose) The model is Cluster patients; Share parameters within clusters. The sampling distribution is

$$p(y_i \mid \theta_j^*, \rho_n) = \mathcal{N}(\theta_j^*, \Sigma)$$

The partition prior is

$$p(\rho_n \mid x^n) \propto \prod_j c(S_j) g(x_j^*)$$

The similarity function is Multivariate normal model.

Covariate-Based vs Augmented Response

An augmented-response model treats (x, y) jointly, so both covariates and responses influence the induced similarity. A covariate-based partition model instead lets the partition prior depend on x and models y conditionally within clusters. The distinction is substantive: joint modeling can make clustering sensitive to arbitrary scaling or misspecification of the covariate density, whereas PPMx keeps the similarity mechanism explicit and preserves a conditional interpretation for the response model.

Chapter 5

Latent Feature Models and the Indian Buffet Process (IBP)

5.1 Infinite Limit Construction

Key Idea: Used vs Unused Features

Let $m_k = \sum_{i=1}^{N} z_{ik}$ be the occupancy of feature k and let $K_+ = \#\{k : m_k > 0\}$ be the number of represented features. Because feature labels have no intrinsic meaning, reorder the columns so that $m_k > 0$ for $k \leq K_+$ and $m_k = 0$ otherwise. In the finite beta–Bernoulli model, this separates the likelihood into K_+ structurally distinct occupied columns and $K - K_+$ identical empty columns. Keeping the combinatorial multiplicity of the empty columns is essential when taking $K \to \infty$.

Algebraic Simplification

The required limit follows from the gamma-function identities

$$\frac{\Gamma(N + 1 + \alpha/K)}{\Gamma(1 + \alpha/K)} = \prod_{j=1}^{N} \left(j + \frac{\alpha}{K}\right) \quad \text{and} \quad \frac{\Gamma(m_k + \alpha/K)}{\Gamma(\alpha/K)} = \left(\frac{\alpha}{K}\right) \prod_{j=1}^{m_k - 1} \left(j + \frac{\alpha}{K}\right).$$

Each occupied column contributes one factor of order K^{-1}; the number of ways to choose and order the occupied columns contributes the compensating power of K.

Limiting Distribution ($K \to \infty$)

Writing $H_N = \sum_{j=1}^{N} j^{-1}$, the limiting probability of an equivalence class $[Z]$ of binary matrices under column permutation is

$$P([Z]) = \frac{\alpha^{K_+}}{\prod_h K_h!} e^{-\alpha H_N} \prod_{k=1}^{K_+} \frac{(N - m_k)!(m_k - 1)!}{N!}.$$

Here K_h is the number of columns having binary history h, so $\prod_h K_h!$ corrects for repeated columns. The factor α^{K_+} controls feature creation, while $e^{-\alpha H_N}$ normalizes the growing collection. The expression depends on rows only through column occupancies and histories, which also establishes row exchangeability.

5.2 Indian Buffet Process (IBP)

The Indian buffet process is an exchangeable distribution on binary feature matrices with finitely many active entries in every row and no fixed upper bound on the number of columns. In its sequential construction, the first observation activates

$$K_1 \sim \text{Poisson}(\alpha)$$

features. Observation i independently reuses an existing feature k with probability

$$\frac{m_k}{i}$$

and activates a Poisson(α/i) number of previously unseen features. Here m_k is the number of preceding rows containing feature k. The sequential description is only a construction; after columns are regarded as unlabelled, the resulting rows are exchangeable.

5.2.1 Posterior Inference

For an existing column, a Gibbs update combines the IBP conditional prior with the likelihood. If $L_{ik}(a)$ denotes the likelihood with $z_{ik} = a$, then

$$\Pr(z_{ik} = 1 \mid Z_{-(ik)}, X) = \frac{m_{-i,k} L_{ik}(1)}{m_{-i,k} L_{ik}(1) + (N - m_{-i,k}) L_{ik}(0)}.$$

This follows from the finite beta–Bernoulli approximation

$$P(z_{ik} = 1 \mid Z_{-i,k}) = \frac{m_{-i,k} + \alpha/K}{N + \alpha/K}$$

by taking $K \to \infty$. New features for row i are proposed according to a Poisson(α/N) prior and accepted or jointly sampled using their marginal likelihood contribution. In nonconjugate models this step commonly requires auxiliary variables or reversible-jump proposals.

5.2.2 Properties of IBP

Every row has marginal feature count

$$K_i \sim \text{Poisson}(\alpha),$$

so the expected number of nonzero entries in an N-row matrix is $N\alpha$. The entries are dependent after feature probabilities are integrated out, so the total number of nonzeros

should not in general be asserted to be Poisson. The number of represented columns does have the exact law

$$K_+ \sim \text{Poisson}(\alpha H_N)$$

because the numbers of new columns introduced by successive rows are independent Poisson variables. Thus $\mathbb{E}K_+ = \alpha H_N \sim \alpha \log N$.

Example 5.2.1. With $\alpha = 2$ and $N = 100$, a row contains two features on average, the matrix contains 200 active entries on average, and $\mathbb{E}K_+ = 2H_{100} \approx 10.37$. Most of those roughly ten columns are sparse, while a few are reused by many rows; this is precisely the regime in which a finite factor model with a preselected rank is awkward.

5.3 Two-Parameter Extension

An additional mass parameter β can be introduced through

$$\pi_k \sim \text{Beta}\left(\frac{\alpha\beta}{K}, \beta\right)$$

so that observation i selects an existing feature with probability

$$\frac{m_k}{\beta + i - 1}$$

and introduces a Poisson$\{\alpha\beta/(\beta + i - 1)\}$ number of new ones. Relative to the one-parameter IBP, β changes the degree of feature reuse without changing the basic interpretation of α as the marginal number of features per row.

Beta Process Connection

The de Finetti mixing measure behind the IBP is a beta process. Draw

$$B \sim \text{BP}(c, B_0), \qquad B = \sum_{k \geq 1} p_k \delta_{\omega_k},$$

and then draw $X_i \mid B \sim \text{BeP}(B)$. Conditional on B, the feature indicators are independent Bernoulli(p_k) variables. Thus the beta process supplies a shared random collection of feature probabilities and the Bernoulli process turns those probabilities into the rows of Z.

Stick-Breaking Construction for Beta Process

One useful ordered construction draws $\mu_k \sim \text{Beta}(\alpha, 1)$ and sets

$$\pi_k = \prod_{j=1}^{k} \mu_j \,.$$

Unlike DP stick weights, the π_k are inclusion probabilities and need not sum to one. An observation can therefore select several atoms independently, which distinguishes a feature allocation from a mixture allocation.

Conjugacy

Beta–Bernoulli conjugacy gives

$$B \sim \mathrm{BP}(c, B_0), \quad X_i \sim \mathrm{BeP}(B) \quad \Longrightarrow \quad B \mid X_{1:n} \sim \mathrm{BP}\left(c+n, \frac{cB_0 + \sum_{i=1}^{n} X_i}{c+n}\right).$$

At an observed atom, this is ordinary beta–Bernoulli updating; the diffuse part of B_0 continues to generate previously unseen features.

5.4 IBP from the Beta–Bernoulli Process

Integrating B out of $B \sim \mathrm{BP}(c, B_0)$ and $X_i \mid B \sim \mathrm{BeP}(B)$ gives

$$X_{n+1} \mid X_{1:n} \sim \mathrm{BeP}\left(\frac{c}{c+n} B_0 + \frac{1}{c+n} \sum_{i=1}^{n} X_i\right).$$

Consequently an existing feature j is selected with probability $m_{n,j}/(c+n)$, while the number of new features is Poisson$\{c\gamma/(c+n)\}$ when $B_0(\Theta) = \gamma$. The one-parameter IBP is recovered by the corresponding parameter choice and indexing convention.

5.5 Hierarchical Beta Process

For grouped feature allocations, use

$$B \sim \mathrm{BP}(c, B_0), \quad A_j \sim \mathrm{BP}(c', B), \quad X_{ij} \sim \mathrm{BeP}(A_j).$$

The global measure B supplies a common set of feature atoms, while A_j assigns group-specific inclusion probabilities to those atoms. Observations in the same group share A_j, so they are more strongly dependent than observations from different groups, but all groups can reuse the same feature semantics.

5.6 Completely Random Measures (CRM)

A random measure assigns a nonnegative random mass $\mu(A)$ to each measurable set $A \subseteq \Theta$. It is *completely random* when disjoint sets have independent masses:

$$A_1 \cap A_2 = \emptyset \Rightarrow \mu(A_1) \perp \mu(A_2).$$

This independence is stronger than independent increments along an ordering; it applies to every finite collection of pairwise disjoint measurable sets.

5.6.1 Lévy–Khintchine Representation

Under the usual regularity conditions, a CRM decomposes into deterministic and fixed-atom parts plus a Poissonian component. Suppressing the first two for clarity, write

$$\mu = \sum_{k=1}^{\infty} w_k \delta_{\theta_k} ,$$

where (w_k, θ_k) are the points of a Poisson process with Lévy intensity $\nu(dw, d\theta)$. The integrability condition $\int (1 \wedge w)\nu(dw, d\theta) < \infty$ on bounded regions ensures finite random mass even though infinitely many small jumps may be present.

5.7 Poisson Point Process

For a Poisson point process with intensity ν, the count in a measurable region A satisfies

$$N(A) \sim \text{Poisson}(\nu(A)) ,$$

and counts on disjoint regions are independent. Conditional on $N(A) = k$, the k points are i.i.d. from the normalized restriction $\nu(\cdot \cap A)/\nu(A)$. These two facts provide the simulation and thinning rules used in CRM constructions.

5.8 Gamma Process

A gamma process with rate b and base measure cH has Lévy intensity

$$\nu(dw, d\theta) = cw^{-1}e^{-bw} dw\, H(d\theta) .$$

The singularity w^{-1} creates infinitely many small positive atoms, while the exponential factor controls large jumps. For measurable A, the mass $\mu(A)$ has a gamma distribution whose shape is proportional to $H(A)$.

5.8.1 Connection to Dirichlet Process

If $0 < H(\Theta) < \infty$, normalization removes the random total mass and gives

$$G = \frac{\mu}{\mu(\Theta)} \sim \text{DP}(\alpha, H/H(\Theta)),$$

with $\alpha = cH(\Theta)$ under this parameterization. Independent gamma increments become Dirichlet proportions after division by their sum; this is the finite-dimensional reason the normalized gamma process is a DP.

5.9 Beta Process

For a beta process $B \sim \mathrm{BP}(c, H)$, one common homogeneous parameterization uses

$$\nu(dw, d\theta) = cw^{-1}(1-w)^{c-1}dw\, H(d\theta)\,, \qquad 0 < w < 1.$$

The atom masses are therefore valid feature probabilities. They are not mixture weights and are not normalized, so several atoms may be active in the same Bernoulli-process draw.

Beta Process → Bernoulli Process → IBP

Given

$$B = \sum_{k=1}^{\infty} w_k \delta_{\theta_k}\,,$$

draw $z_{ik} \mid B \sim \mathrm{Bernoulli}(w_k)$ independently over k and set $X_i = \sum_k z_{ik}\delta_{\theta_k}$. Marginalizing B induces the IBP law on the binary matrix $Z = (z_{ik})$. This relationship is analogous to gamma-process normalization producing a DP, but here no normalization occurs.

5.10 Interpretation Across Processes

The gamma and beta processes serve different inferential roles. Gamma-process atoms have masses in $(0, \infty)$ and become mixture probabilities after normalization, producing a Dirichlet process. Beta-process atoms have masses in $(0, 1)$ and act directly as independent inclusion probabilities. Conditional Bernoulli-process draws generate binary features, and integrating out their shared beta process produces the IBP distribution on Z.

Posterior Inference for IBP Models

For an occupied feature, the generic conditional update is

$$P(z_{ik} = 1 \mid Z_{-i,k}, X) \propto \frac{m_{-i,k}}{N} \cdot P(X \mid Z)\,,$$

with the corresponding $z_{ik} = 0$ likelihood in the normalizing denominator. Row i also proposes $\mathrm{Poisson}(\alpha/N)$ new features. Columns are conditionally independent given their beta-process masses, while rows are exchangeable after those masses are integrated out; confusing these two levels leads to incorrect independence claims.

5.11 Example: Linear Gaussian Latent Feature Model

For observations $x_i \in \mathbb{R}^d$, a linear latent-feature model is

$$x_i \sim \mathcal{N}(z_i A, \sigma^2 I), \qquad \text{or in matrix form} \qquad X = ZA + \epsilon.$$

Here z_i is a binary row vector and the rows of A are feature loadings. Although Z has countably many potential columns, a finite sample activates only finitely many, so the matrix product is almost surely well defined. With a Gaussian prior on A, the loadings can be integrated out to obtain a collapsed likelihood for Gibbs updates of Z.

Key Properties of IBP Models

An IBP prior combines an unbounded feature vocabulary with finitely many active features per observation. Its row-exchangeable law makes it suitable for i.i.d. observational units, while the nonexclusive columns permit overlapping latent causes rather than a single cluster assignment.

Extension: Feature Modeling across multiple groups

Suppose articles are collected from several news outlets and may express multiple latent topics. A beta–Bernoulli model can simultaneously discover the topic vocabulary and the subset used by each article:

$$B \sim \mathrm{BP}(c, H), \qquad B = \sum_{k=1}^{\infty} w_k \delta_{\theta_k}, \qquad z_{ik} \mid B \sim \mathrm{Bernoulli}(w_k).$$

Here θ_k denotes topic k, w_k is its population inclusion probability, and $z_{ik} = 1$ records its presence in article i.

Limitation of the Vanilla Beta Process

The standard model assumes $P(z_{ik} = 1) = w_k$ for every article. It therefore forces all outlets to have the same topic prevalence, even when they share only the meanings of the topics. A hierarchical beta process separates these two notions. The global measure

$$B_0 \sim \mathrm{BP}(c_0, H), \qquad B_0 = \sum_{k=1}^{\infty} p_k \delta_{\theta_k}$$

defines the shared atoms. Outlet j receives

$$B_j \mid B_0 \sim \mathrm{BP}(c_1, B_0), \qquad B_j = \sum_{k=1}^{\infty} w_{jk} \delta_{\theta_k}, \qquad w_{jk} \sim \mathrm{Beta}\{c_1 p_k, c_1(1 - p_k)\},$$

and article i from outlet j draws $X_{ji} \mid B_j \sim \mathrm{BeP}(B_j)$. The atoms remain globally aligned, while w_{jk} allows topic prevalence to vary by outlet. Large c_1 shrinks group prevalences toward p_k; small c_1 permits stronger heterogeneity.

5.12 From IBP to Count-Valued Feature Models

The beta–Bernoulli construction

$$B \sim \mathrm{BP}(c, H), \quad X_i \mid B \sim \mathrm{BeP}(B)$$

produces binary indicators $z_{ik} \in \{0, 1\}$. In RNA sequencing and document analysis, however, the observation is a count rather than mere presence. A count-valued feature model must represent both whether a factor is available and how intensely it contributes.

5.13 Negative Binomial Factor Analysis (NBFA)

Let X_{vj} be the count of gene v in cell j. Negative-binomial factor analysis uses

$$X_{vj} \sim \mathrm{NB}\left(\sum_k \phi_{vk}\theta_{kj},\ p_j\right),$$

where ϕ_{vk} is the loading of gene v on factor k, θ_{kj} is the activity of that factor in sample j, and p_j controls dispersion. The additivity of the negative binomial permits the augmentation

$$X_{vj} = \sum_k X_{vjk}, \quad X_{vjk} \sim \mathrm{NB}(\phi_{vk}\theta_{kj}, p_j),$$

which allocates each observed count among latent factors and yields tractable conditional updates under gamma priors.

5.14 Double Feature Allocation (DFA)

Double feature allocation represents overlapping latent structure on both sides of a relational matrix. In an electronic-health-record application, patients may exhibit several diseases and each disease may affect several symptoms. Unlike biclustering with exclusive row and column clusters, DFA permits both memberships to overlap.

5.15 Latent Disease Interpretation

The observed patient–symptom matrix is generated through two unobserved relations: patient-to-disease membership and disease-to-symptom effects. This factorization distinguishes a latent condition from its manifestations and allows different diseases to produce the same symptom.

5.16 Patient–Disease (PD) Model

An IBP prior on

$$A = (\alpha_{ik}) \in \{0, 1\}^{n \times \infty} \sim \mathrm{IBP}(m)$$

lets column k represent a latent disease and sets $\alpha_{ik} = 1$ when patient i has that disease. Although the prior admits countably many diseases, only finitely many are represented among finitely many patients.

5.17 Symptom–Disease (SD) Model

Conditional on the active diseases, let

$$C = (\gamma_{lk}) \in \{-1, 0, 1\}^{p \times K} \quad \text{with} \quad C \sim \text{Categorical}(\pi), \qquad \pi \sim \text{Dirichlet}(\phi).$$

The sign of γ_{lk} records whether disease k suppresses, does not affect, or promotes symptom l:

$$\gamma_{lk} = \begin{cases} -1 & \text{disease } k \text{ decreases symptom } l \\ 0 & \text{no relation} \\ +1 & \text{disease } k \text{ increases symptom } l \end{cases}$$

5.18 Sampling Model

For patient i and symptom l, the response $y_{il} \in \{-1, 0, +1\}$ records negative, absent, or positive association. A categorical sampling model is

$$y_{il} \sim \text{Categorical}\left(Me^{\alpha_i^\top W^- \gamma_l^- + \eta_l^-},\ M,\ Me^{\alpha_i^\top W^+ \gamma_l^+ + \eta_l^+}\right)$$

where W^- and W^+ are sign-specific weight matrices, η_l^- and η_l^+ are symptom-specific offsets, and M normalizes the three category weights. This form separates latent mechanisms that suppress and promote a symptom rather than forcing them to have equal magnitude.

5.19 Scalability Challenges

Standard IBP MCMC repeatedly updates the allocation matrix A, usually one row, column, or entry at a time. The resulting serial dependence can yield slow mixing, while storing likelihood contributions for many patients and a growing number of active feature columns creates substantial memory cost. The central scalability problem is therefore not merely large n: it is the combination of a large binary matrix, an unknown number of columns, and label nonidentifiability across independent computations.

5.20 Consensus Monte Carlo

Consensus Monte Carlo partitions the observations, runs subset posterior samplers independently, and combines their draws. This is straightforward for an identified Euclidean parameter, but an IBP feature matrix is invariant to column permutations. Two workers may learn the same latent feature under different labels, or learn different numbers of features, so entrywise averaging of their binary matrices has no statistical meaning.

5.21 Modified Consensus Monte Carlo

A modified construction places a common set of anchor observations in every subset. Feature columns are matched by comparing their activation patterns and associated parameters on these anchors; only after this matching step are subset summaries combined. The anchors make distributed computation more scalable, but they do not remove approximation error: weakly identified or nearly duplicate features can still be matched incorrectly, so sensitivity to the anchor size and matching threshold should be reported.

Chapter 6

Polya Tree Priors

6.1 Prior Construction

The DP constructs a random probability measure through random atoms and the IBP constructs an unbounded binary feature allocation. A Pólya tree instead assigns random probability recursively to a nested sequence of measurable partitions. It can be viewed as an indefinitely refined random histogram: the coarse levels determine large-scale mass allocation and deeper levels encode local detail.

6.2 Ingredients of a Polya Tree

A Pólya tree is specified by a recursive partition of the sample space and a random branching probability at every split. Multiplying the branch probabilities along a path gives the mass of the corresponding cell.

6.3 Recursive Partition Tree

Let $(\Omega, \mathcal{B})$ be a measurable space and let

$$\Pi = \{\pi_m : m = 0, 1, 2, \dots\}$$

be nested partitions. The root is $\pi_0 = \{\Omega\}$, each π_{m+1} refines π_m, and in the binary construction every cell splits into two disjoint children.

6.3.1 Binary Indexing

Let $E = \{0, 1\}$, E^m be the binary strings of length m, and $E^* = \bigcup_{m \geq 0} E^m$. At level m the partition is

$$\pi_m = \{B_\epsilon : \epsilon \in E^m\}.$$

The string records the path from the root, and every parent is partitioned by its two children:

$$B_\epsilon = B_{\epsilon 0} \cup B_{\epsilon 1}, \quad B_{\epsilon 0} \cap B_{\epsilon 1} = \emptyset$$

6.4 Polya Tree Definition

Assign an independent branching probability to every internal node,

$$Y_\epsilon := G(B_{\epsilon 0} \mid B_\epsilon) \sim \text{Beta}(\alpha_{\epsilon 0}, \alpha_{\epsilon 1}).$$

The child masses satisfy

$$G(B_{\epsilon 0}) = G(B_\epsilon) Y_\epsilon, \quad G(B_{\epsilon 1}) = G(B_\epsilon)(1 - Y_\epsilon)$$

and hence a depth-m cell has mass

$$G(B_\epsilon) = \prod_{j=1}^{m} Y_{\epsilon_1 \cdots \epsilon_{j-1}}^{1-\epsilon_j} (1 - Y_{\epsilon_1 \cdots \epsilon_{j-1}})^{\epsilon_j}$$

with the empty prefix denoting the root.

Example 6.4.1. Partition $[0, 1]$ at $1/2$, then split each half at its midpoint. With $Y_\varnothing \sim \text{Beta}(2, 2)$, $Y_0 \sim \text{Beta}(3, 1)$, and $Y_1 \sim \text{Beta}(1, 3)$, the prior mean masses of the four quarters are $3/8, 1/8, 1/8, 3/8$. After observing counts $(6, 2, 1, 5)$, conjugacy gives $Y_\varnothing \mid X \sim \text{Beta}(10, 8)$, $Y_0 \mid X \sim \text{Beta}(9, 3)$, and $Y_1 \mid X \sim \text{Beta}(2, 8)$. This small calculation shows that a Polya tree updates locally: only branching variables on an observation's path are affected.

6.5 Tailfree Processes

The conditional branch probability is

$$Y_\epsilon = \frac{G(B_{\epsilon 0})}{G(B_\epsilon)}$$

whenever the denominator is positive. A prior is tailfree when the collections of branching variables at successive levels,

$$\{Y_\emptyset\}, \{Y_0, Y_1\}, \{Y_{00}, Y_{01}, Y_{10}, Y_{11}\}, \ldots$$

are independent. Independent beta branches therefore make the Pólya tree a tailfree process.

6.6 Dirichlet Process as a Special Case

Suppose

$$Y_\epsilon \sim \text{Beta}(\alpha_{\epsilon 0}, \alpha_{\epsilon 1})$$

with projectively consistent parameters

$$\alpha_\epsilon = \alpha_{\epsilon 0} + \alpha_{\epsilon 1}, \quad \alpha_\epsilon = c\, H(B_\epsilon)$$

for a probability measure H. The resulting finite-dimensional masses agree with those of

$$G \sim \mathrm{DP}(cH)$$

on every partition generated by the tree. Thus a DP is a specially coordinated Pólya tree; arbitrary beta parameters generally define a different prior.

6.6.1 Consistency Property

Suppose that the parameters of a parent equal the sum of its children,

$$\alpha_0 = \alpha_{00} + \alpha_{01}, \quad \alpha_1 = \alpha_{10} + \alpha_{11}\,.$$

Dirichlet aggregation then implies

$$(G(B_{00}), G(B_{01}), G(B_{10}), G(B_{11})) \sim \mathrm{Dirichlet}(\alpha_{00}, \alpha_{01}, \alpha_{10}, \alpha_{11})$$

Iterating this argument gives

$$(G(B_\epsilon) : |\epsilon| = m) \sim \mathrm{Dirichlet}(\alpha_\epsilon : |\epsilon| = m)$$

6.7 Interpretation

A Pólya tree is a distribution over probability measures built from recursive beta splits. Its node-specific parameters provide local prior control: one region may be tightly concentrated around the centering measure while another remains diffuse.

6.8 Absolute Continuity of Polya Trees

A classical sufficient condition for absolute continuity is that the concentration parameters increase sufficiently quickly with depth. For the symmetric choice

$$\alpha_{\epsilon_1 \cdots \epsilon_m} = cm^2,$$

Kraft's theorem gives $G \ll G_0$ almost surely. At level m the split obeys

$$Y_\epsilon \sim \mathrm{Beta}(\alpha_m, \alpha_m), \quad \alpha_m = cm^2$$

and therefore has mean $1/2$ and variance of order m^{-2}. The summability of these fine-scale fluctuations prevents persistent singular mass from forming. This differs sharply from a Dirichlet process draw,

$$G = \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k}\,,$$

which is almost surely discrete. Absolute continuity does not by itself imply a smooth density; stronger growth or mixing over partitions is used when smoothness is required.

6.9 Effect of Parameters

The scale c controls variability around the centering measure G_0. Increasing c concentrates each split more tightly around its prior mean; decreasing it produces rougher, more variable random densities. The rate of growth across tree levels additionally controls fine-scale smoothness.

6.10 Prior Centering

A Pólya tree prior is written

$$G \sim PT(\Pi, \mathcal{A})$$

where Π is the recursive partition and $\mathcal{A} = \{(\alpha_{\epsilon 0}, \alpha_{\epsilon 1})\}$ contains the beta parameters. Centering at G_0 means

$$\mathbb{E}[G] = G_0$$

6.11 Centering Method 1: Quantile Partition

Choose symmetric branch priors,

$$\alpha_{\epsilon 0} = \alpha_{\epsilon 1}$$

and construct the partition from G_0-quantiles: split first at the median, then at the first and third quartiles, and continue recursively. Every child then has half its parent's G_0-mass, so symmetric beta means reproduce the centering probabilities.

6.12 Centering Method 2: Match Split Probabilities

For a fixed partition Π, choose parameters satisfying

$$\frac{\alpha_{\epsilon 0}}{\alpha_{\epsilon 0} + \alpha_{\epsilon 1}} = G_0(B_{\epsilon 0} \mid B_\epsilon)$$

or, equivalently,

$$\alpha_{\epsilon 0} \propto G_0(B_{\epsilon 0} \mid B_\epsilon), \quad \alpha_{\epsilon 1} \propto G_0(B_{\epsilon 1} \mid B_\epsilon)$$

with a common node-specific proportionality constant. This separates the prior mean, determined by the ratio, from local precision, determined by the sum of the two parameters.

6.13 Why This Centers the Prior

By definition,

$$Y_\epsilon = \frac{G(B_{\epsilon 0})}{G(B_\epsilon)}$$

Then

$$G(B_{\epsilon 0}) = G(B_\epsilon) Y_\epsilon$$

Taking conditional expectation gives

$$\mathbb{E}[G(B_{\epsilon 0}) \mid G(B_\epsilon)] = G(B_\epsilon)\mathbb{E}[Y_\epsilon]$$

Because

$$\mathbb{E}[Y_\epsilon] = \frac{\alpha_{\epsilon 0}}{\alpha_{\epsilon 0} + \alpha_{\epsilon 1}} = G_0(B_{\epsilon 0} \mid B_\epsilon)$$

it follows that

$$\mathbb{E}[G(B_{\epsilon 0}) \mid G(B_\epsilon)] = G(B_\epsilon)G_0(B_{\epsilon 0} \mid B_\epsilon)$$

Recursing down the tree yields

$$\mathbb{E}[G(B_{\epsilon_1 \cdots \epsilon_m})] = G_0(B_{\epsilon_1 \cdots \epsilon_m})$$

6.14 Posterior Inference

For $x_1, \ldots, x_n \mid G \overset{iid}{\sim} G$, every observation traverses one path through the partition. At node ϵ, let

$$n_{\epsilon 0} = \#\{x_i \in B_{\epsilon 0}\}, \quad n_{\epsilon 1} = \#\{x_i \in B_{\epsilon 1}\}\,.$$

Since $Y_\epsilon \sim \text{Beta}(\alpha_{\epsilon 0}, \alpha_{\epsilon 1})$, beta–binomial conjugacy acts independently at every node and gives

$$Y_\epsilon \mid x_{1:n} \sim \text{Beta}(\alpha_{\epsilon 0} + n_{\epsilon 0}, \alpha_{\epsilon 1} + n_{\epsilon 1})$$

6.15 Posterior Predictive Sampling

To generate x_{n+1}, begin at the root and recursively choose a child using a draw from the updated branch law

$$Y_\epsilon \mid x_{1:n} \sim \text{Beta}(\alpha^*_{\epsilon 0}, \alpha^*_{\epsilon 1})\,.$$

After truncating at a sufficiently deep leaf B_ϵ, sample from the centering distribution restricted to that cell, $x_{n+1} \sim G_0(\cdot \mid B_\epsilon)$. Alternatively, branch decisions can be integrated out using posterior beta means; this produces the exact finite-depth posterior predictive distribution.

6.16 Marginal Distribution

The marginal likelihood

$$p(x_1, \ldots, x_n) = \int \prod_{i=1}^{n} G(x_i)\, dP(G)$$

factorizes over nodes because the branch probabilities are independent a priori. At each node, integrating its beta-distributed split against the left and right traversal counts yields a beta–binomial ratio. Multiplying these ratios over all occupied nodes gives a tractable recursive likelihood; two observations interact precisely along the portion of their paths that they share.

6.17 Polya Tree: Marginal Distribution

The marginal density after integrating out G:

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} G_0(x_i) \prod_{j=2}^{n} \prod_{m=1}^{m^*(x_j)} \frac{\alpha^*_{\epsilon_m(x_j)}}{\alpha_{\epsilon_m(x_j)}} \cdot \frac{\alpha_{\epsilon_m 0(x_j)} + \alpha_{\epsilon_m 1(x_j)}}{\alpha^*_{\epsilon_m 0(x_j)} + \alpha^*_{\epsilon_m 1(x_j)}}$$

Here $\epsilon_m(x_j)$ denotes the level-m cell containing x_j, $m^*(x_j)$ is the first level at which it is isolated, and α^* denotes posterior-updated parameters. The factor $\prod_i G_0(x_i)$ supplies the centering contribution, while the remaining ratios adjust it according to the observations' shared paths.

6.18 Mixture of Polya Trees (MPT)

A single fixed partition can imprint artificial discontinuities on posterior density estimates. A mixture of Pólya trees randomizes a centering or partition parameter,

$$G \mid \theta \sim PT(\mathcal{A}, G_{0,\theta}), \quad \theta \sim p(\theta)\,.$$

Averaging over θ moves the cell boundaries and reduces these artifacts. Smoothness is a property of the resulting mixture, not of every conditional Pólya-tree draw.

6.19 Motivation: Joint Modeling

For a time-to-event outcome T and a longitudinal response Y, the target is their joint law. The two factorizations

$$[T, Y] = [Y][T \mid Y] = [T][Y \mid T]$$

are mathematically equivalent but lead to different models. Placing a Pólya tree on $[T]$ and a regression model on $[Y \mid T]$ offers a convenient way to represent heterogeneous event-time distributions without forcing the longitudinal component to carry that flexibility.

6.20 Multivariate Polya Tree

To center a multivariate Pólya tree at

$$G_0 = \mathcal{N}_p(\mu, \Sigma), \quad \Sigma = UU^\top\,,$$

first form the univariate Gaussian-quantile cells

$$B_0(m, k) = (\Phi^{-1}((k-1)/2^m), \Phi^{-1}(k/2^m)]$$

and take their Cartesian products,

$$B_0(m, \mathbf{k}) = \prod_{j=1}^{p} B_0(m, k_j)\,.$$

The affine image

$$B(m, \mathbf{k}) = \{\mu + Uz : z \in B_0(m, \mathbf{k})\}$$

is aligned with the geometry of G_0 and partitions $\mathbb{R}^p$ into 2^{mp} equal-G_0-probability cells at depth m.

6.21 Rubbery Polya Tree

A rubbery Pólya tree reduces boundary discontinuities by correlating nearby splits. For example, the hierarchy

$$Y_{00} \sim \text{Beta}(\alpha_2, \alpha_2), \quad Z_{00} \mid Y_{00} \sim \text{Binomial}(\delta_{00}, Y_{00})$$

$$Y_{10} \mid Z_{00} \sim \text{Beta}(\alpha_2 + Z_{00}, \alpha_2 + \delta_{00} - Z_{00})$$

uses the auxiliary count Z_{00} to transmit information from one split to its neighbor. After marginalization the beta behavior is retained, but adjacent split probabilities are positively correlated, producing smoother random densities.

6.22 Linear Dependent Tailfree Process (LDTP)

A linear dependent tailfree process turns the Pólya tree into a conditional density model by replacing every fixed branch probability with

$$G_x(B_{e0} \mid B_e) = Y_{x,e0} \quad \text{where} \quad Y_{x,e0} = \frac{\exp(x^\top \beta_{e0})}{1 + \exp(x^\top \beta_{e0})}.$$

Because every node has its own coefficient, changing x can alter location, scale, skewness, modality, and tail mass rather than only a single moment.

6.23 Why LDTP Helps

Under Gaussian regression,

$$Y_i \mid x_i \sim \mathcal{N}(\mu(x_i), \sigma^2(x_i)),$$

the conditional shape remains Gaussian even if both moments vary. An LDTP instead uses $Y_i \mid x_i \sim G_{x_i}$ and lets the complete conditional law change with x_i. Deep-node coefficients require shrinkage because the corresponding cells contain little data.

6.24 LDTP Case Study: Growth Curve Analysis

Let Y_i be immunoglobulin concentration and x_i age. The model

$$Y_i = m(x_i) + \epsilon_i, \quad \epsilon_i \sim G_{x_i},$$

with an LDTP prior on G_x, separates a broad growth curve from age-dependent residual skewness, tails, and modality. Posterior conditional quantiles are more informative than a mean curve when the distribution changes shape with age.

6.25 Dependent Multivariate Polya Trees

The objective is a stochastic family

$$\mathcal{G} = \{G_x : x \in \mathcal{X}\}$$

such that every G_x is marginally a Pólya tree and nearby covariate values produce dependent distributions. Replace each scalar branch variable by a process,

$$Y_\epsilon \to \{Y_{\epsilon,x} : x \in \mathcal{X}\},$$

whose one-dimensional marginals satisfy

$$Y_{\epsilon,x} \sim \text{Beta}(\alpha_\epsilon, \alpha_\epsilon)$$

and whose dependence is controlled through

$$\{Y_{\epsilon,x}\}_{x\in\mathcal{X}} \sim \text{MBP}(\alpha_0, \alpha_1, Q).$$

The kernel Q determines how quickly association decays across covariate values.

6.26 Multivariate Beta Process (MBP)

A multivariate beta process produces beta-distributed variables $\{Y_x : x \in \mathcal{X}\}$ with dependence across x. The construction shares gamma increments in the ratio

$$Y = \frac{G^0}{G^0 + G^1}, \quad G^0, G^1 \sim \Gamma.$$

Two covariate values are correlated to the extent that their ratios reuse the same gamma-process mass.

6.27 MBP Construction

Let G be a gamma process on an augmented space and let

$$Q = \{q_x : x \in \mathcal{X}\}$$

be a collection of nonnegative kernels. Define

$$S_x^0 = \{(t, y) : 0 < y < \alpha_0 q_x(t)\}, \quad S_x^1 = \{(t, y) : -\alpha_1 q_x(t) < y < 0\}$$

and set

$$Y_x = \frac{G(S_x^0)}{G(S_x^0) + G(S_x^1)}.$$

The gamma ratio gives the required beta marginal. Overlap between the regions for x and x' makes the corresponding gamma masses, and hence the branch probabilities, dependent.

6.28 Finite-Dimensional Representation for MCMC

For observed covariates $x_1, \ldots, x_m$, introduce branch indicators

$$Z_i \mid Y_{x_i} \sim \text{Bernoulli}(Y_{x_i})$$

and a DP representation for the latent gamma-process points,

$$(\xi_{ij}, \zeta_{ij}) \sim \text{DP}(\nu_{x_1,\ldots,x_m}) .$$

For each i, locate the first atom falling in S_{x_i} and set $Z_i = \mathbb{I}(\zeta_{ij} > 0)$. This representation replaces direct sampling of a dependent vector of beta variables by a DP Pólya-urn calculation and therefore yields a tractable MCMC update.

6.29 Mixture of Polya Trees Revisited

The mixture model

$$G \mid \theta \sim PT(\mathcal{A}, G_{0,\theta}), \quad \theta \sim p(\theta)$$

averages over partition locations or centering parameters. Its posterior predictive density is smoother and less sensitive to any one arbitrary set of tree boundaries.

6.30 Polya Trees in Joint Modeling

The factorization $[T, Y] = [T][Y \mid T]$ permits a Pólya-tree prior for the event-time marginal and a regression model for the longitudinal response conditional on event time. This avoids a rigid parametric survival distribution while retaining an interpretable conditional model for Y.

6.31 Summary of Extensions

These extensions address distinct limitations of a standard Pólya tree. Mixing over centering or partition parameters reduces fixed-boundary artifacts. An LDTP makes the conditional distribution predictor dependent. More general dependent trees preserve beta marginals while coupling branch probabilities across covariate values, and the multivariate beta-process construction supplies both that dependence and a DP-based computational representation.

6.32 Alternative Construction via MCMC

At a fixed node, introduce the branch indicators

$$Z_i \in \{0, 1\}, \quad i = 1, \ldots, m ,$$

where one and zero denote the left and right children. Conditional on the covariate-specific split,

$$Z_i \mid Y_{x_i} \sim \text{Bernoulli}(Y_{x_i}) .$$

The vector $Z = (Z_1, \ldots, Z_m)$ is sufficient for the next level of tree recursion, so the sampler can target these indicators without explicitly drawing every Y_{x_i}.

6.33 DP Representation

Let

$$S = \bigcup_{i=1}^{m} S_{x_i}, \quad S_{x_i} = S^0_{x_i} \cup S^1_{x_i}$$

and draw

$$D_{x_1,\ldots,x_m} \sim DP(\nu_{x_1,\ldots,x_m})$$

Conditional on that random measure, generate latent points

$$(\xi_{ij}, \zeta_{ij}) \in S, \quad (\xi_{ij}, \zeta_{ij}) \mid D \overset{iid}{\sim} D_{x_1,\ldots,x_m}$$

6.34 Polya Urn Sampling

Writing $w_r = (\xi_r, \zeta_r)$, the Pólya-urn predictive update is

$$w_r \mid w_1, \ldots, w_{r-1} \sim \frac{\nu(S)}{\nu(S) + r - 1}\nu + \frac{1}{\nu(S) + r - 1} \sum_{\ell=1}^{r-1} \delta_{w_\ell}$$

For observation i, let

$$j_i = \inf\{j : (\xi_{ij}, \zeta_{ij}) \in S_{x_i}\}$$

Define the branching indicator

$$Z_i = \mathbb{I}(\zeta_{ij_i} > 0)$$

Thus a hit in $S^0_{x_i}$ sends the observation left and a hit in $S^1_{x_i}$ sends it right.

6.35 Take-Home Message

The MBP representation reduces LDTP computation to a DP Pólya urn over latent points. Sampling those points and their induced branch indicators avoids a high-dimensional direct update of $\{Y_{x_i}\}$ and yields a reusable MCMC primitive at every occupied tree node.

Chapter 7

Bayesian Nonparametric Survival Analysis

7.1 Survival Analysis: Setup

Let T_i be an event time and C_i an independent right-censoring time. The observed variables are

$$Y_i = \min(T_i, C_i), \quad \delta_i = \mathbb{I}(T_i < C_i)\,.$$

Under independent censoring, the likelihood contribution involving the event-time law is

$$\prod_{i=1}^{n} f(Y_i)^{\delta_i} S(Y_i)^{1-\delta_i}$$

The survival, hazard, and cumulative-hazard functions are

$$S(t) = \Pr(T > t), \quad \lambda(t) = \frac{f(t)}{S(t)}, \quad \Lambda(t) = \int_0^t \lambda(u)\,du$$

7.2 Semiparametric Survival Models

7.2.1 Proportional Hazards (PH)

$$\lambda_x(t) = \lambda_0(t)\exp(x^\top \beta)$$

Here λ_0 is an unrestricted baseline hazard and $\exp(x^\top\beta)$ is a multiplicative covariate effect. Consequently the hazard ratio is constant in time:

$$\frac{\lambda_{x_1}(t)}{\lambda_{x_2}(t)} = \exp((x_1 - x_2)^\top \beta)$$

7.2.2 Accelerated Failure Time (AFT)

$$\log T = x^\top \beta + \epsilon$$

Equivalently, covariates rescale time according to

$$S_x(t) = S_0(\exp(-x^\top \beta)t)$$

A BNP extension places a DP-mixture prior on the error law, retaining the time-rescaling interpretation while relaxing the parametric shape of the baseline survival distribution.

7.2.3 Proportional Odds (PO)

$$\frac{S_x(t)}{1 - S_x(t)} = \frac{S_0(t)}{1 - S_0(t)} \exp(-x^\top \beta)$$

The proportional-odds restriction is distinct from proportional hazards and generally permits a time-varying hazard ratio.

7.3 Bayesian Semiparametric PH Models

A Bayesian semiparametric analysis places a prior on the baseline hazard or cumulative hazard,

$$\lambda_0(t) \quad \text{or} \quad \Lambda_0(t)$$

Common choices include gamma-process and beta-process priors on cumulative hazard increments and piecewise-exponential priors on the hazard itself. The posterior then quantifies uncertainty in the entire baseline survival curve, which propagates into covariate-specific predictions.

7.4 Neutral-to-the-Right (NTR) Processes

A random CDF F is NTR if

$$1 - F(t_1),\ \frac{1 - F(t_2)}{1 - F(t_1)}, \dots$$

are independent for every $t_1 < \cdots < t_k$. Equivalently,

$$F(t) = 1 - \exp\{-Y(t)\}$$

where $Y(t)$ is a nondecreasing process with independent increments and plays the role of a random cumulative hazard. This multiplicative-increment structure is well matched to right-censored likelihoods and yields useful conjugacy. The Dirichlet process is a special member of the NTR class under an appropriate construction.

7.5 Fully Nonparametric Survival Regression

The target is the full conditional family

$$\{S(t \mid x) : x \in \mathcal{X}\}$$

rather than a baseline curve modified only by proportional hazards or log-linear time scaling.

7.5.1 ANCOVA-DDP

The model is

$$y_i = \log T_i = x_i^\top \beta + \epsilon_i$$

The error distribution receives the prior

$$\epsilon_i \sim G_{x_i}$$

where $\{G_x\}$ follows a dependent Dirichlet process. Dependence across x borrows strength between nearby covariates, while the marginal error shape may change with x.

7.5.2 Linear Dependent Tailfree Process (LDTP)

An LDTP replaces each Pólya-tree split by

$$G_x(B_{e0} \mid B_e) = \frac{\exp(x^\top \beta_{e0})}{1 + \exp(x^\top \beta_{e0})}$$

and thereby produces a family

$$G = \{G_x : x \in \mathcal{X}\}$$

whose complete conditional density, including its tails and modality, may change with x.

7.6 Why BNP Survival Models

Parametric survival distributions can impose an incorrect hazard shape, while standard semiparametric models retain a fixed low-dimensional form for covariate effects. BNP survival regression places a prior on the full conditional event-time distribution, allowing skewness, multimodality, and effects that vary across time or regions of the response distribution.

7.7 Fully Nonparametric Survival Regression: LDTP (Interpretation)

7.7.1 Interpretation I

The LDTP is fully nonparametric because every node e of the infinite tree has its own coefficient β_{e0}. Covariates enter through the local split

$$G_x(B_{e0} \mid B_e) = \frac{\exp(x^\top \beta_{e0})}{1 + \exp(x^\top \beta_{e0})}$$

at every resolution. A predictor can therefore move mass differently in the center and tails, allowing asymmetric effects that cannot be summarized by one global regression coefficient.

7.7.2 Interpretation II

A regularizing prior is

$$\beta_{e0} \sim \mathcal{N}\left(0, g(X^\top X)^{-1}\right), \quad g = \frac{2n}{c}$$

which shrinks deep-node splits toward one half and limits overfitting where data are sparse. Both the LDTP and ANCOVA-DDP define a nonparametric family $\{G_x\}$. The ANCOVA-DDP obtains smooth densities through a Gaussian mixture; the LDTP directly randomizes recursive conditional probabilities, giving more local control but partition-dependent computation.

7.8 Alternative BNP Construction: DDP-GP

7.8.1 Model Construction

Replace a linear component mean with

$$\theta_h(Z) \sim \mathcal{GP}(\mu_h, C)$$

$$\mu_h(Z_i; \beta_h) = Z_i^\top \beta_h$$

and covariance

$$C(Z_i, Z_\ell) = \sigma_0^2 \exp\left(-\sum_{d=1}^{D} \frac{(Z_{id} - Z_{\ell d})^2}{\lambda_d^2}\right) + \delta_{i\ell} J^2$$

7.8.2 Interpretation

The length scale λ_d determines how rapidly the latent mean changes in covariate direction d, so nearby covariates induce similar component means. The nugget J^2 stabilizes covariance computations and may also represent unstructured component-level variation.

7.9 DDP-GP Model for Survival Data

The observed data satisfy $Y_i = \log(T_i)$; Z_i: covariates; δ_i: censoring indicator. The model is

$$p(y_i \mid Z_i, F) = F_{Z_i}(y_i)$$

The prior is

$$\{F_Z\} \sim \text{DDP-GP}$$

Under right censoring, the likelihood is

$$\mathcal{L}(\theta \mid D_n) = \prod_{i=1}^{n} \left\{ f_{Z_i}(Y_i \mid \theta)^{\delta_i} \left[1 - F_{Z_i}(Y_i \mid \theta)\right]^{1-\delta_i} \right\}$$

This construction means that Observed events $\Rightarrow$ density contribution; Censored observations $\Rightarrow$ survival contribution.

7.10 Comparison of BNP Survival Models

The ANCOVA-DDP is mixture based and has a natural clustering interpretation; the LDTP is tree based and offers explicit local control of distributional shape; the DDP–GP induces smooth covariate dependence through a kernel. The choice should follow the inferential structure: latent subpopulations favor a DDP mixture, local tail effects favor an LDTP, and smooth functional evolution favors a GP-dependent construction.